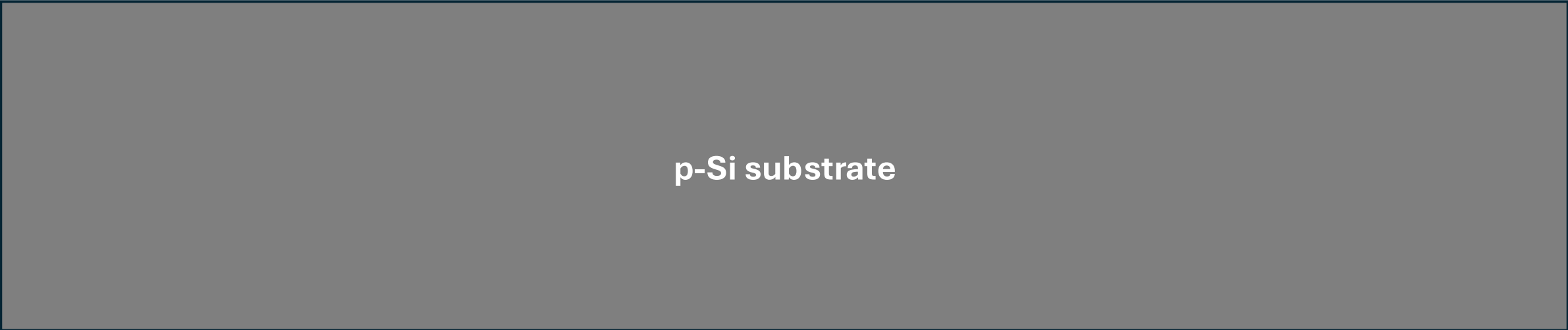


CMOS Fabrication Process Flow

Nitupon Dihingia

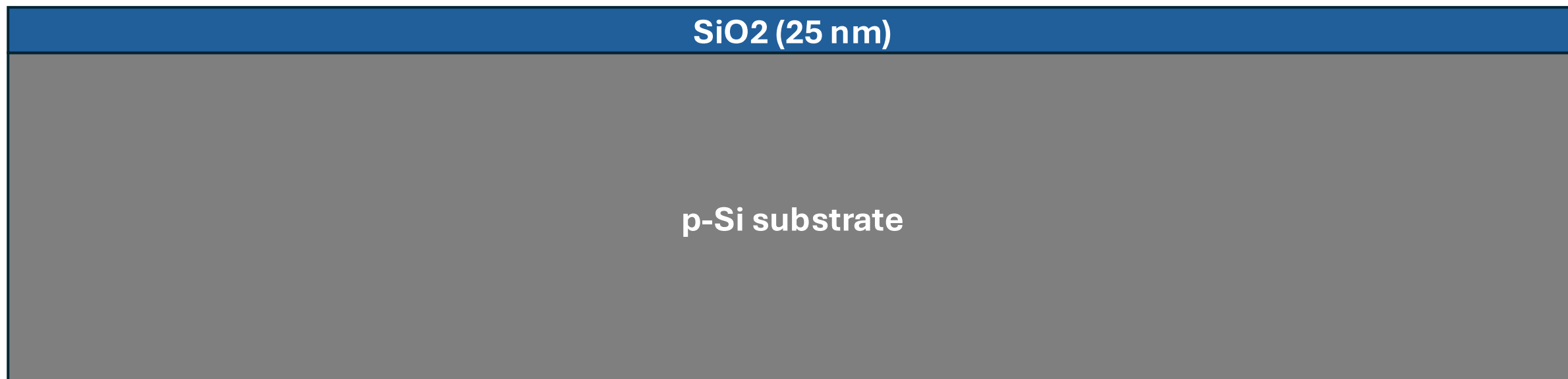
Process Engineer - Litho, Monk Tech

Dimensions not to Scale

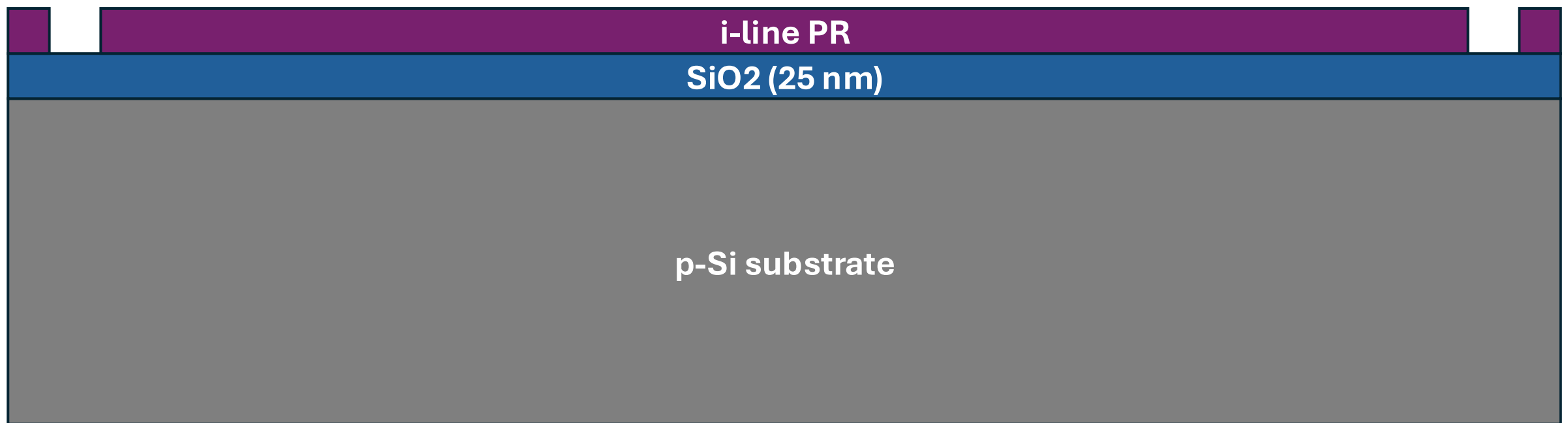


p-Si substrate

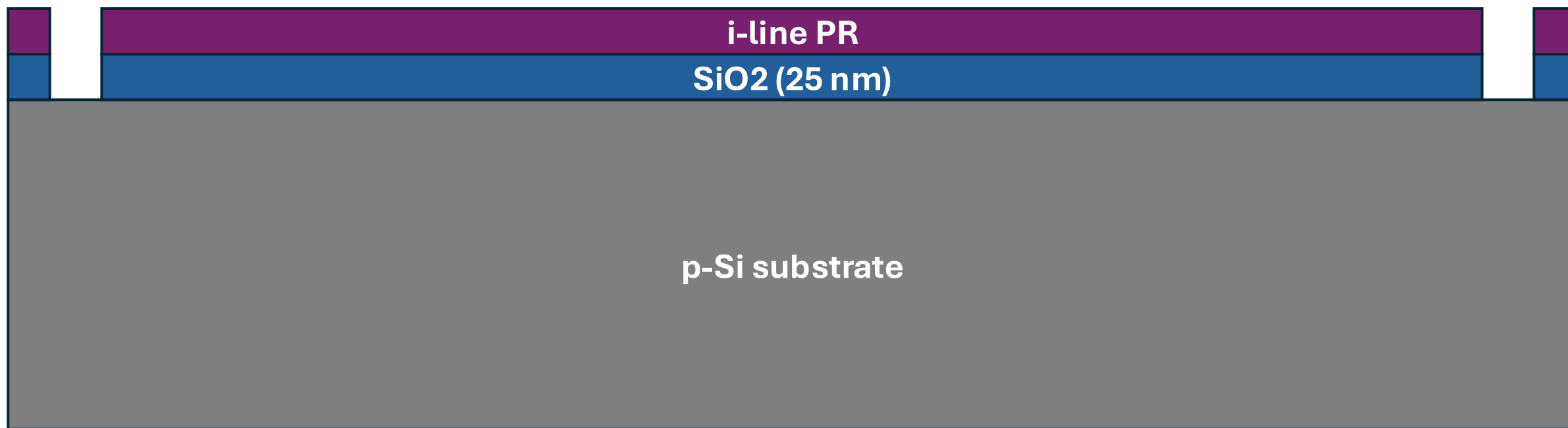
1. Wafer Start



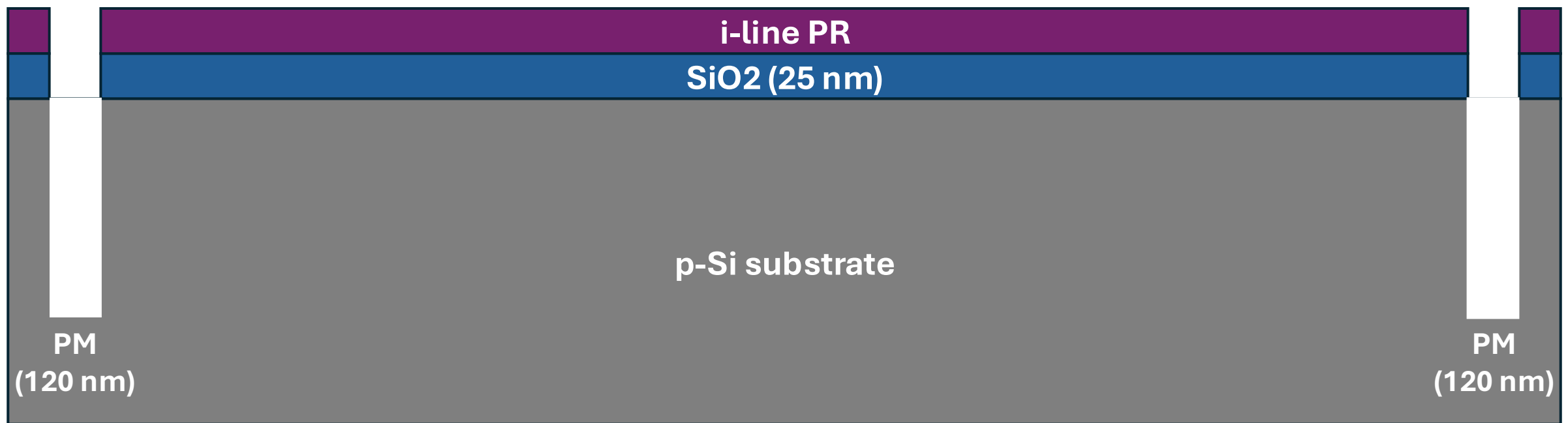
2. SiO2 growth



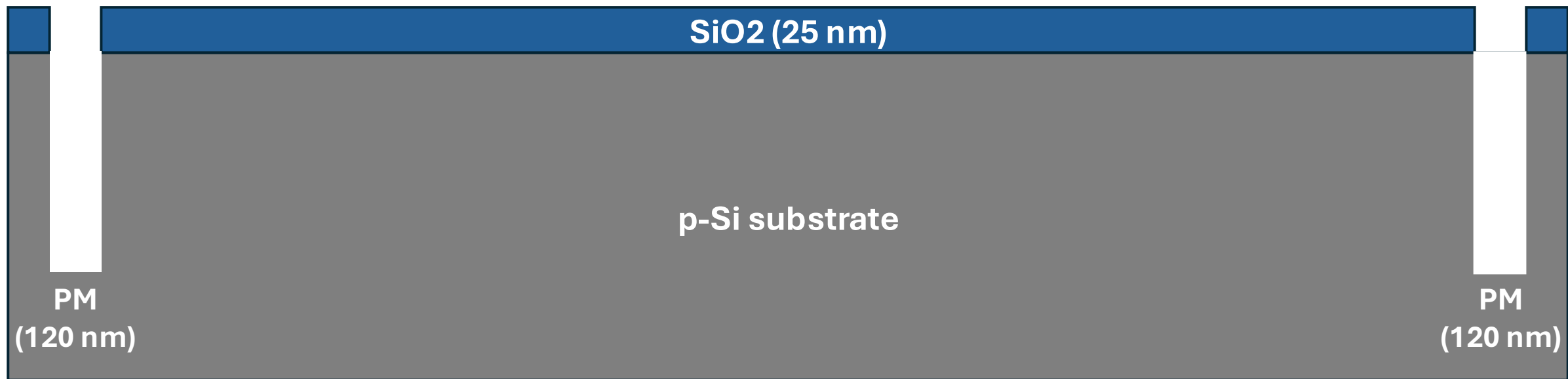
3. Zero Layer Litho



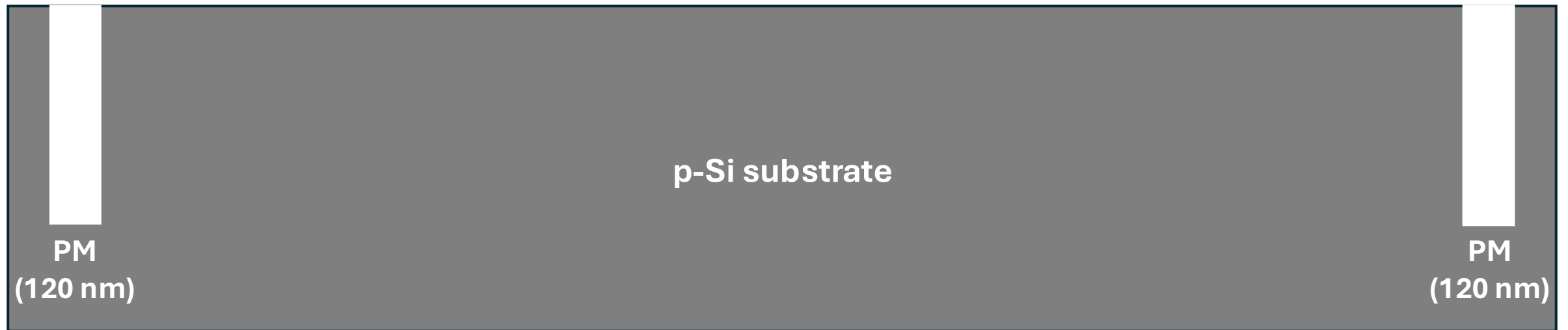
4. SiO2 dry etch



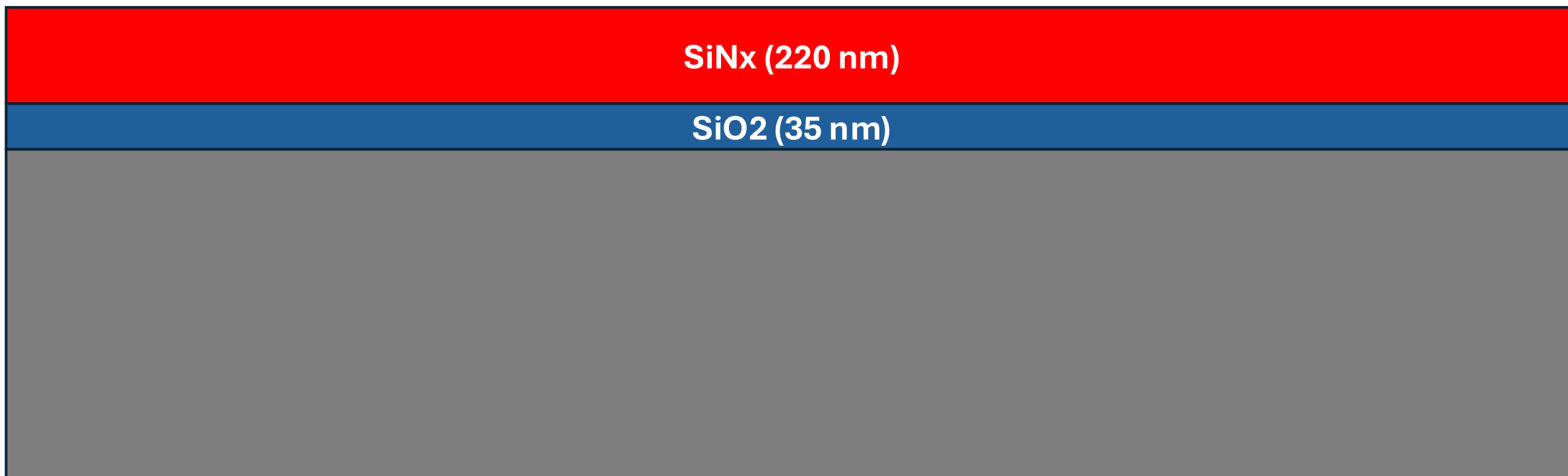
5. PM Marks Dry Etch



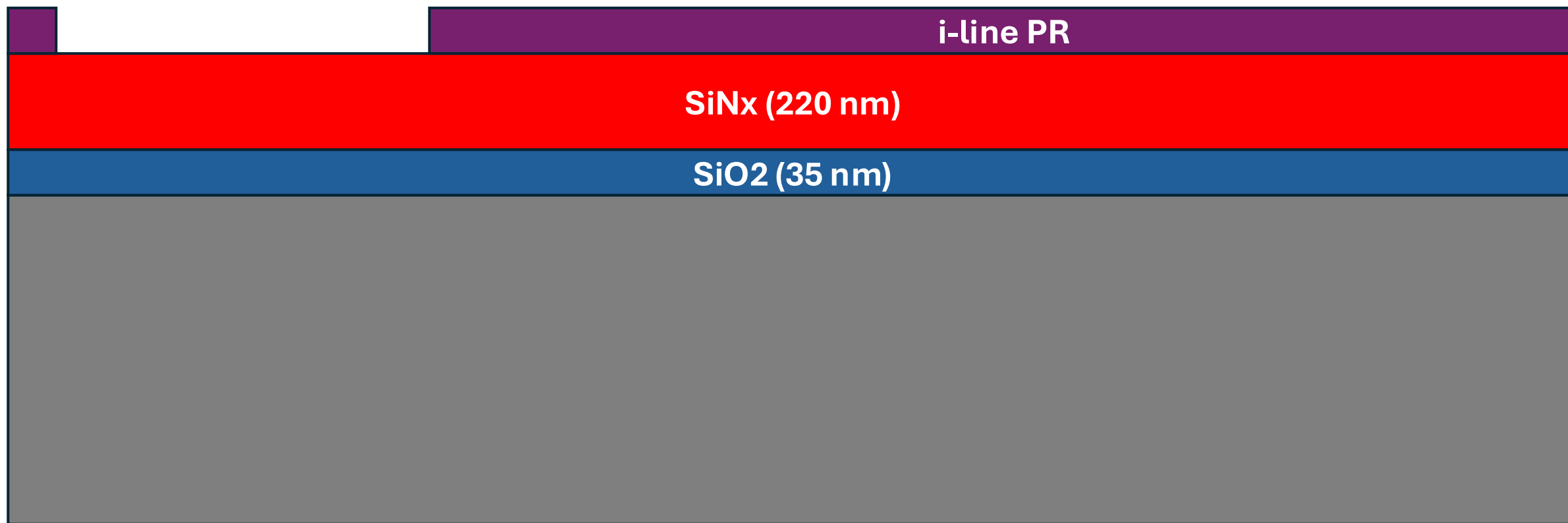
6. PR Strip



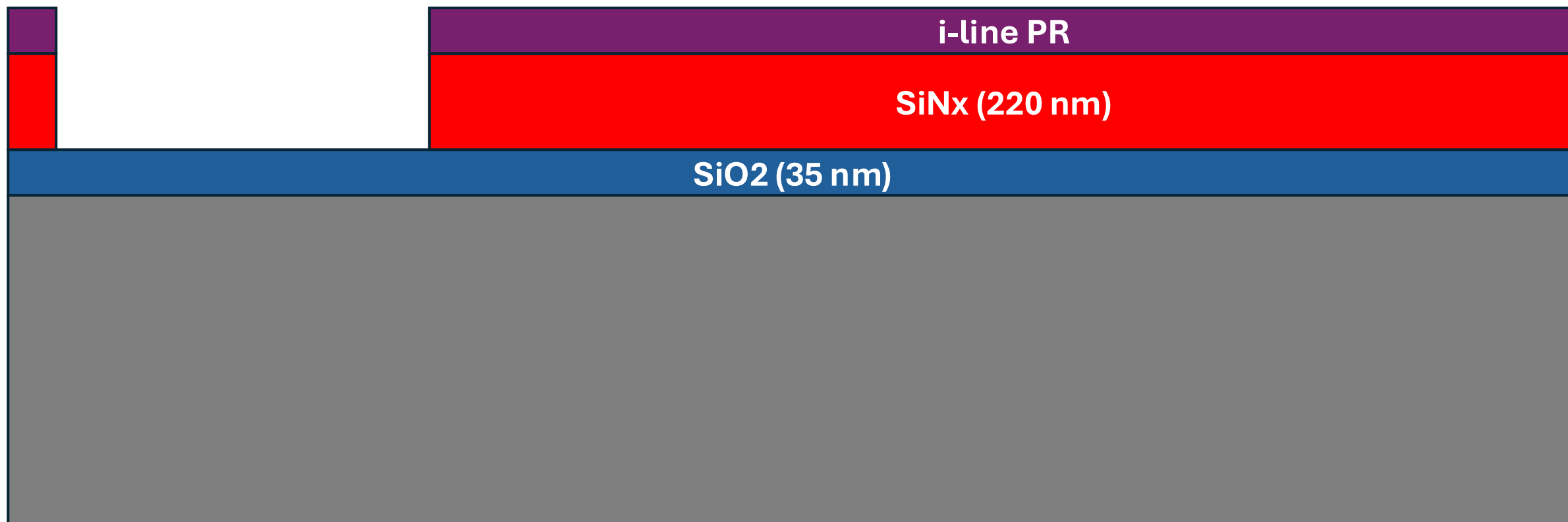
7. SiO₂ Wet Etch



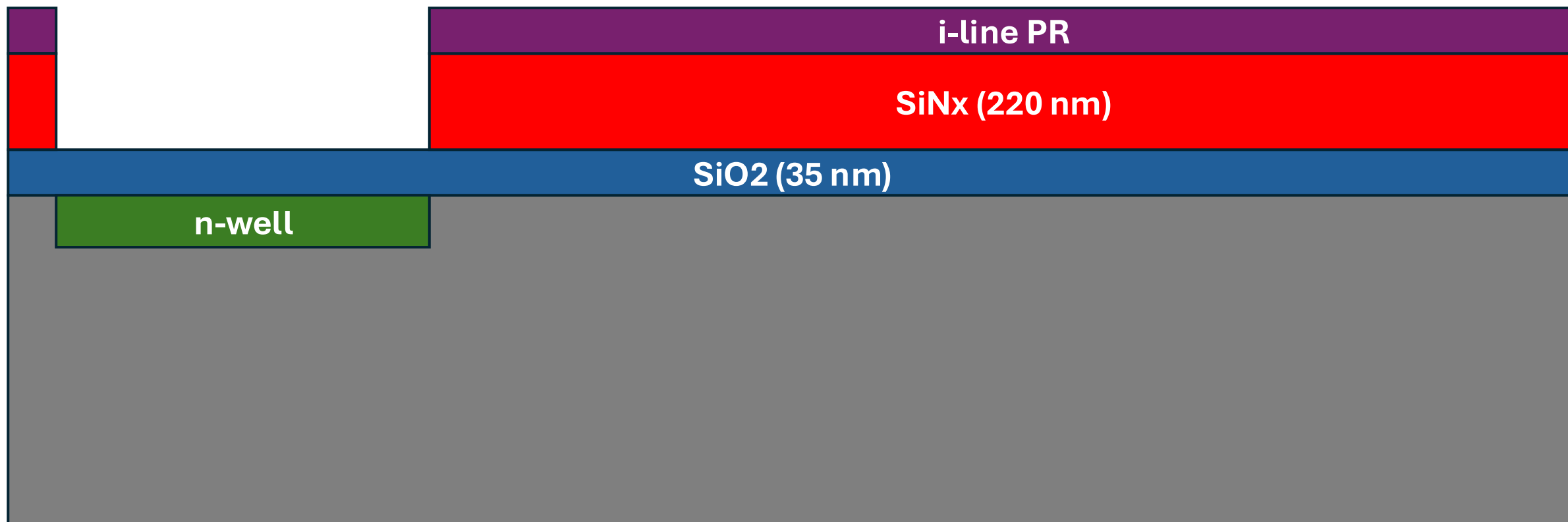
8. Pad Oxide + Nitride Deposition



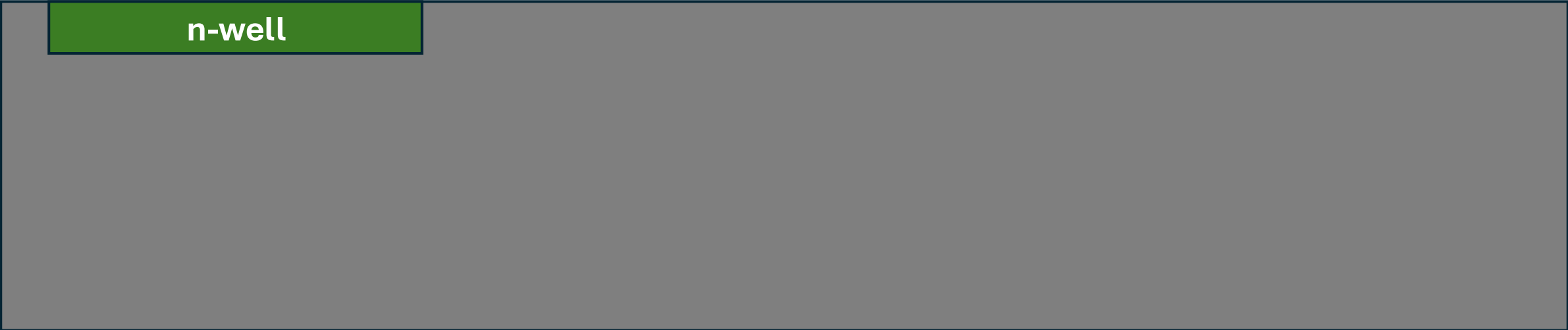
9. n-well Litho



10. Nitride Wet Etch



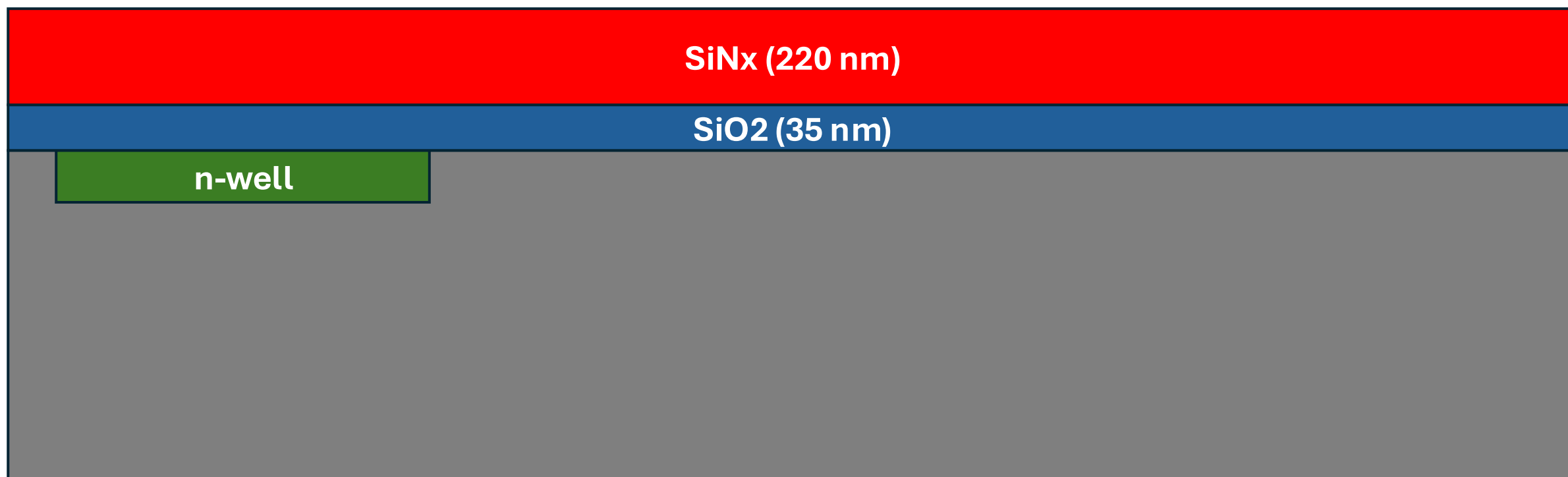
11. n-well Implantation



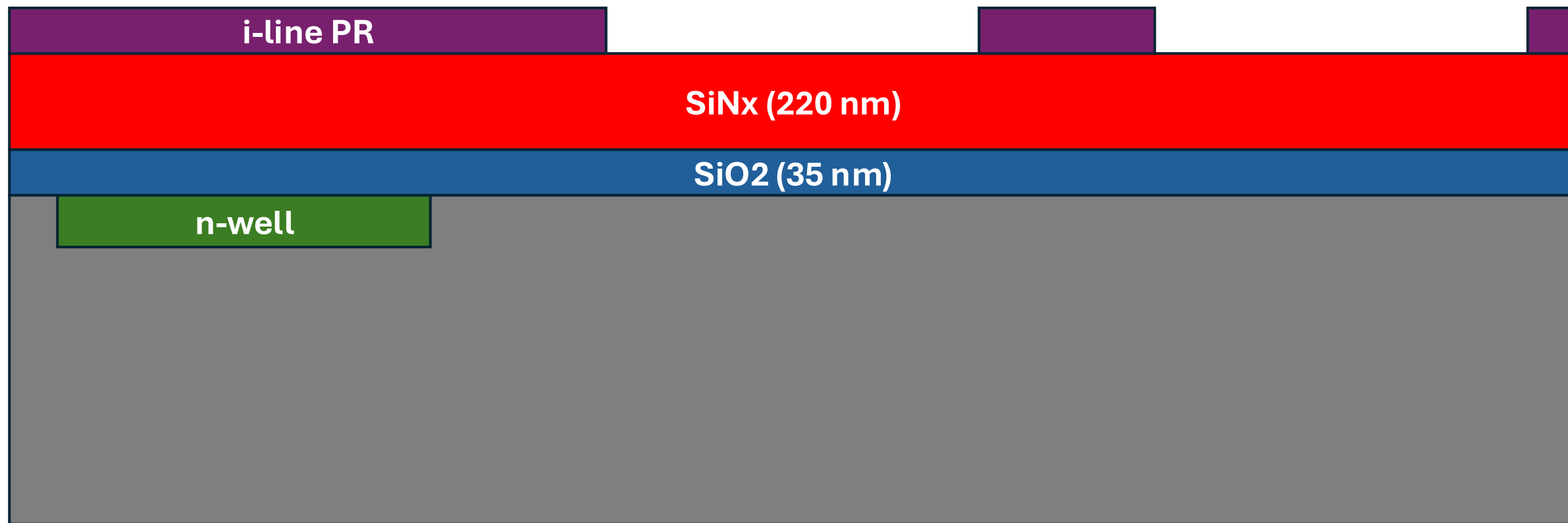
A cross-sectional diagram of a semiconductor device. It shows a gray substrate with a green rectangular region on top, labeled "n-well".

n-well

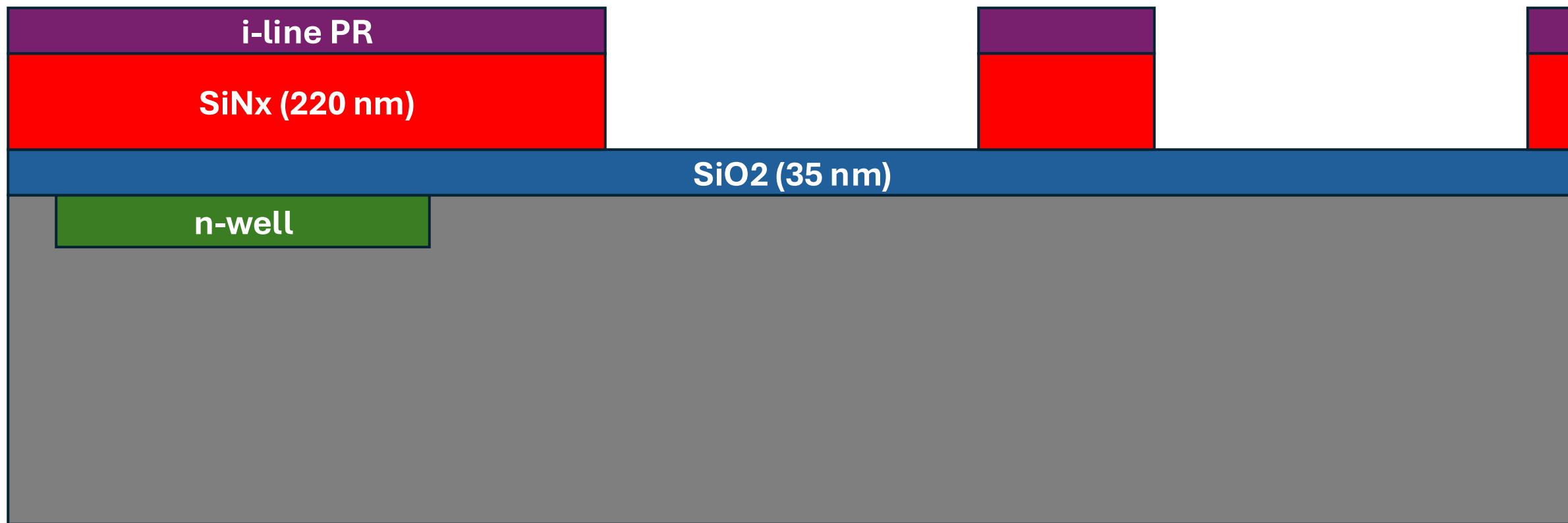
12. PR + Oxide + Nitride Removal



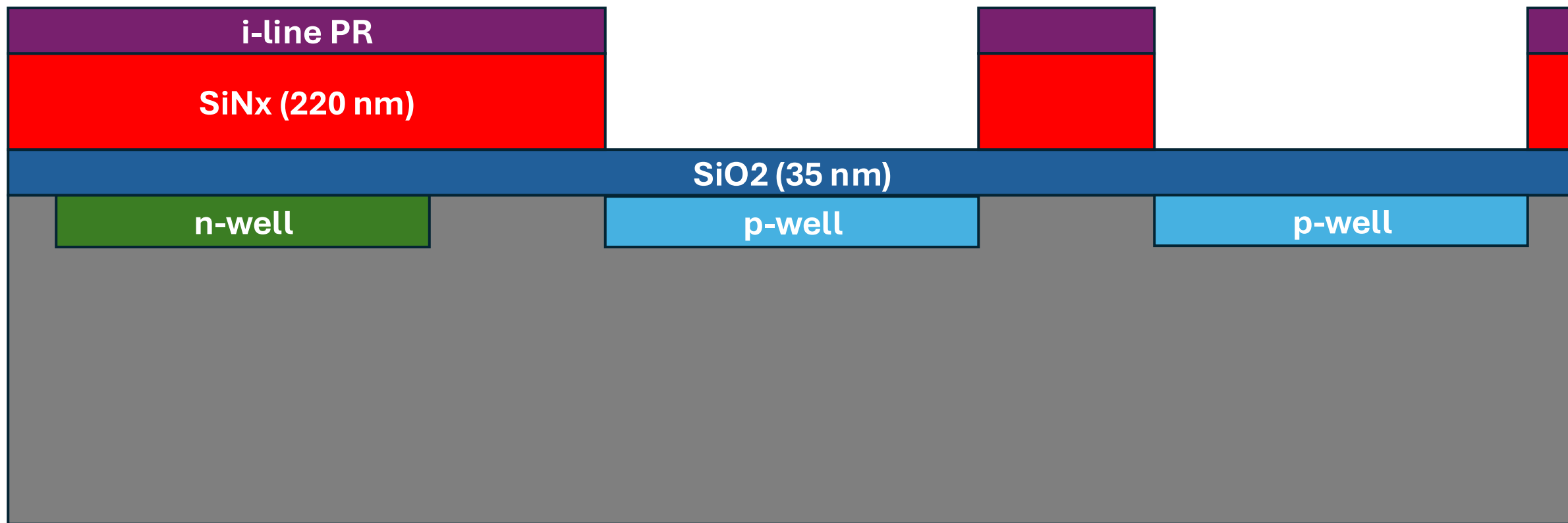
13. Pad Oxide + Nitride Deposition



14. p-well Litho



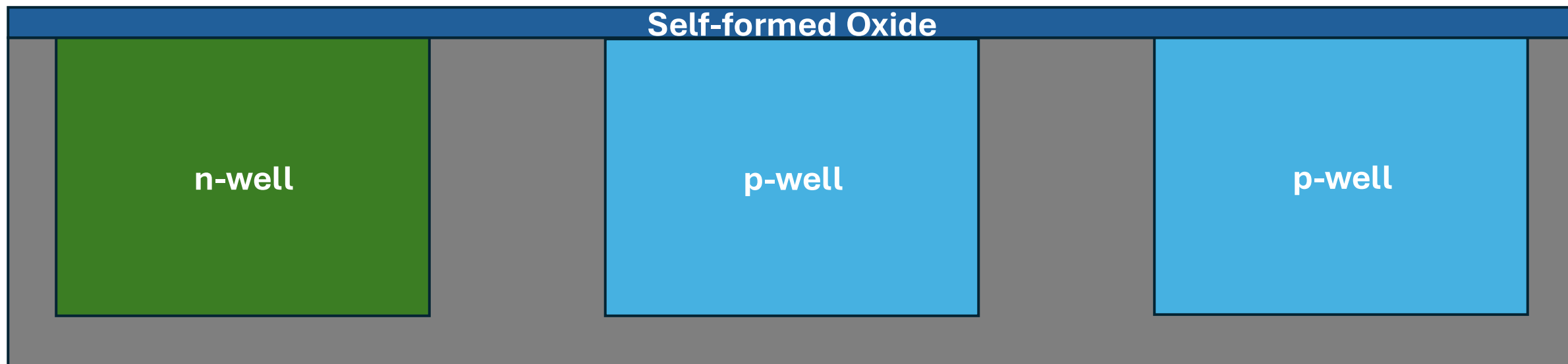
15. Nitride Dry Etch



16. p-well Implantation



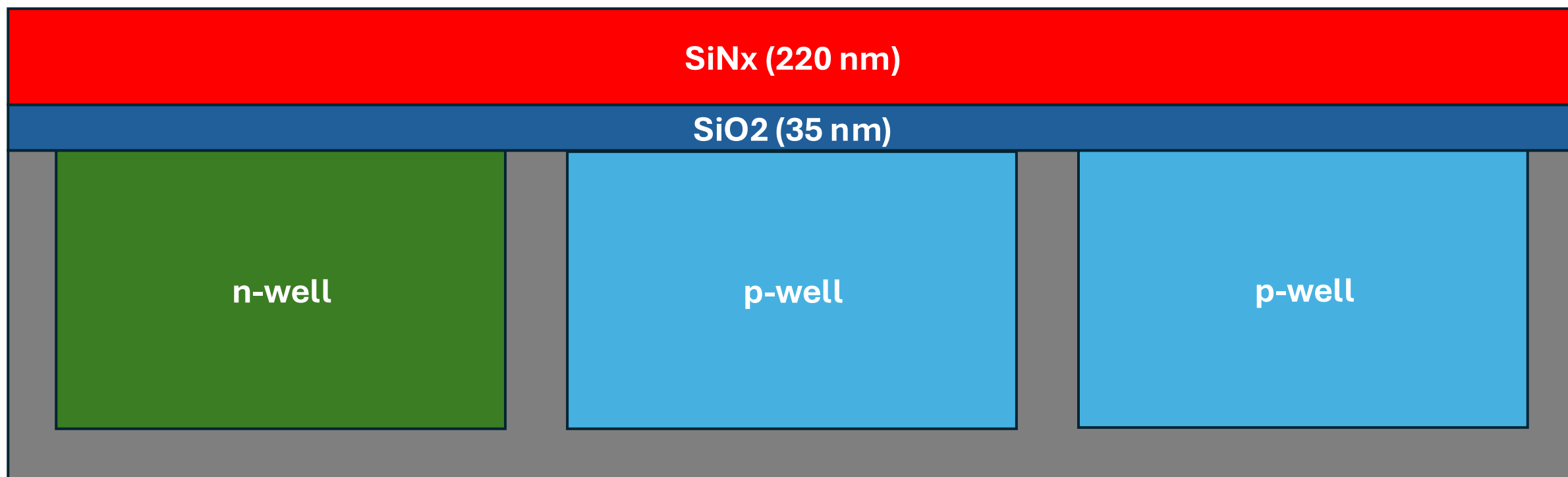
17. PR + Oxide + Nitride Removal



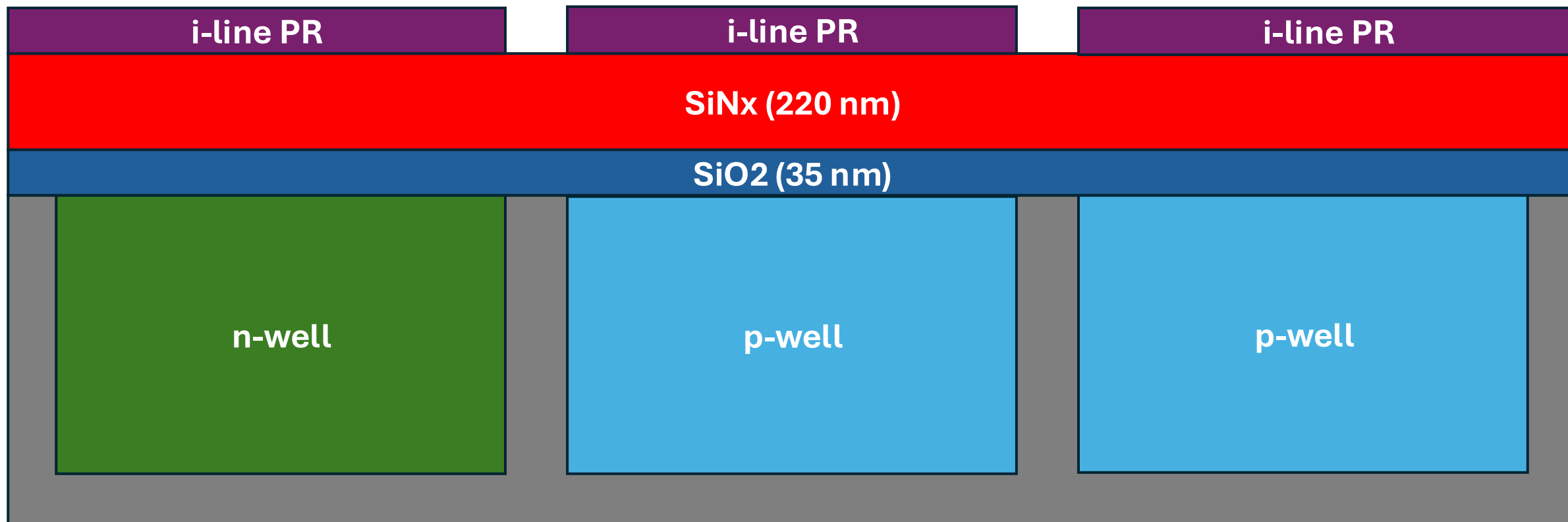
18. Well Drive-In



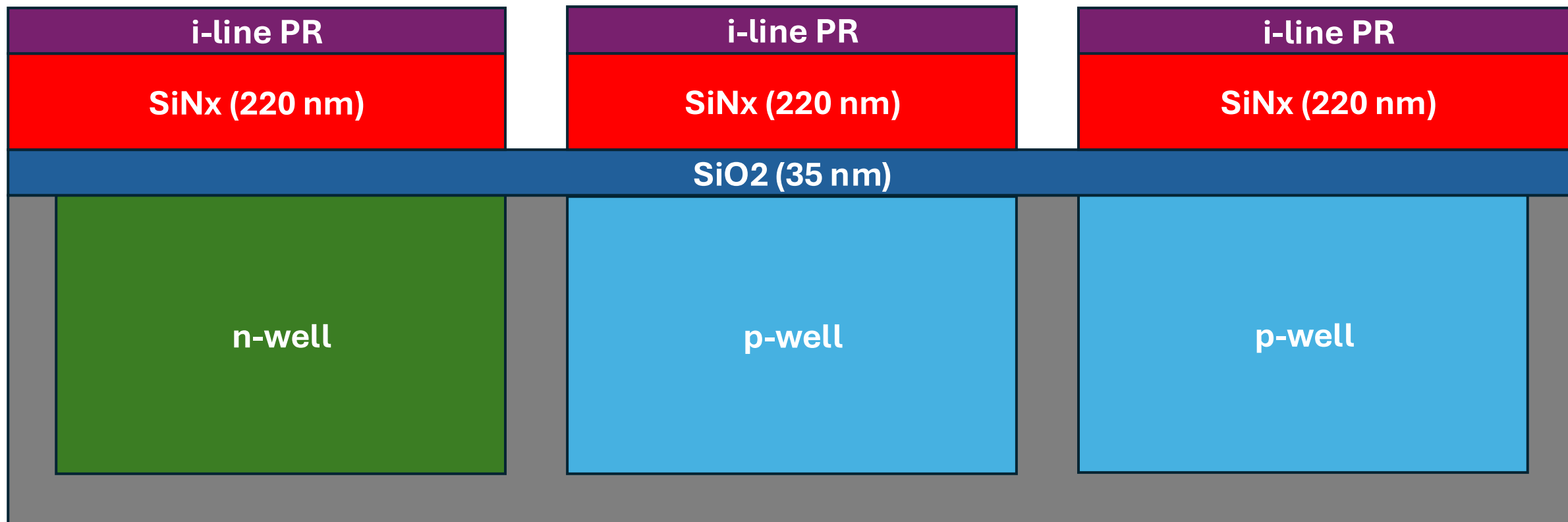
19. Oxide Wet Etch



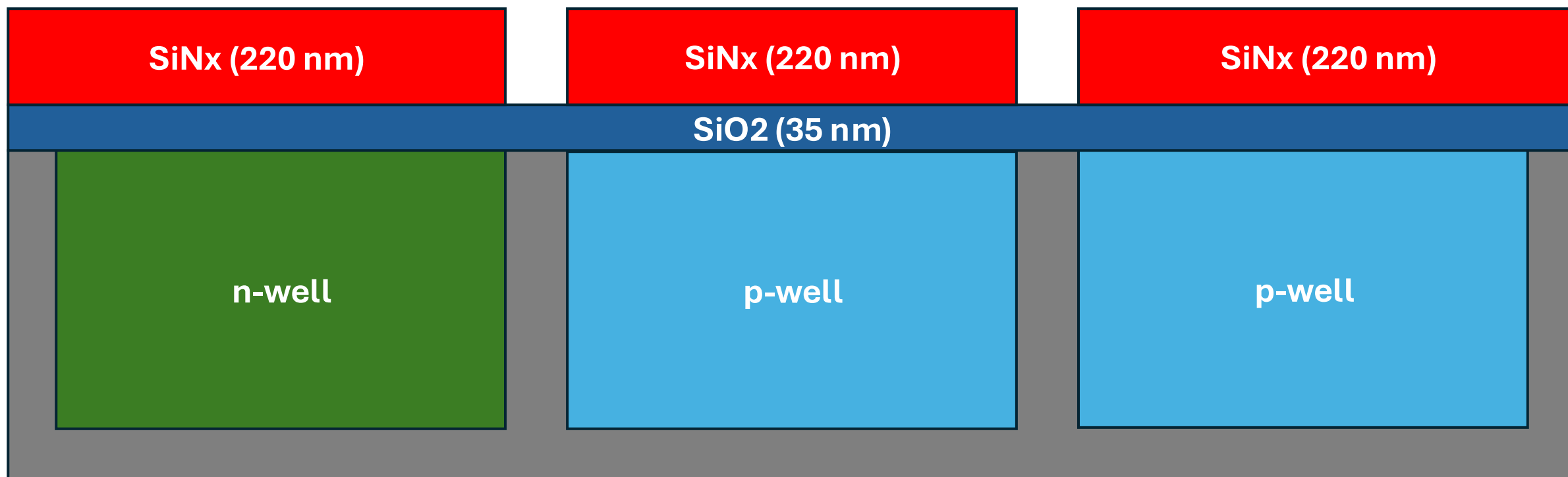
20. Oxide + Nitride Deposition

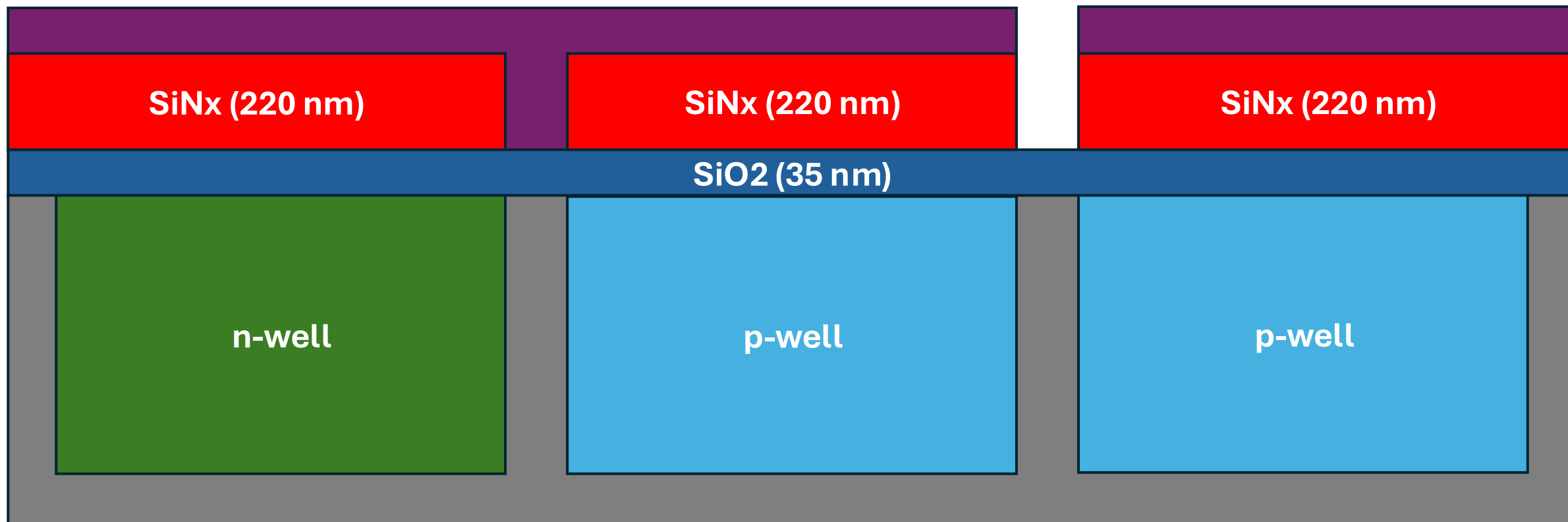


21. Active Area Litho

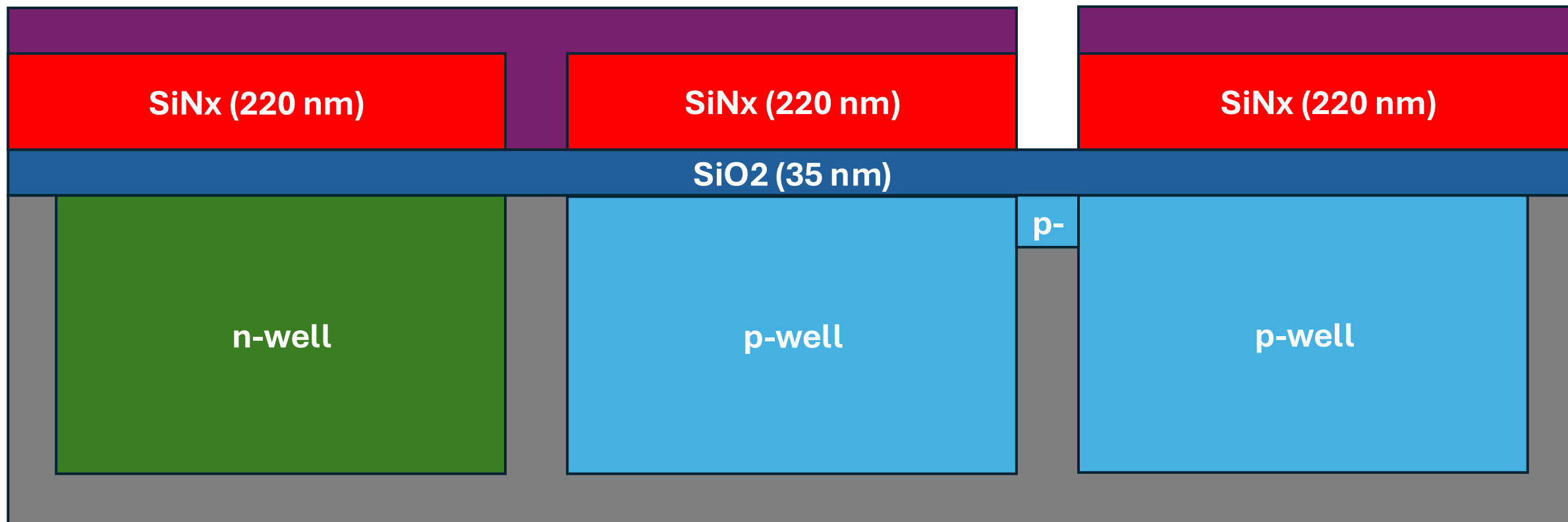


22. Nitride Etch

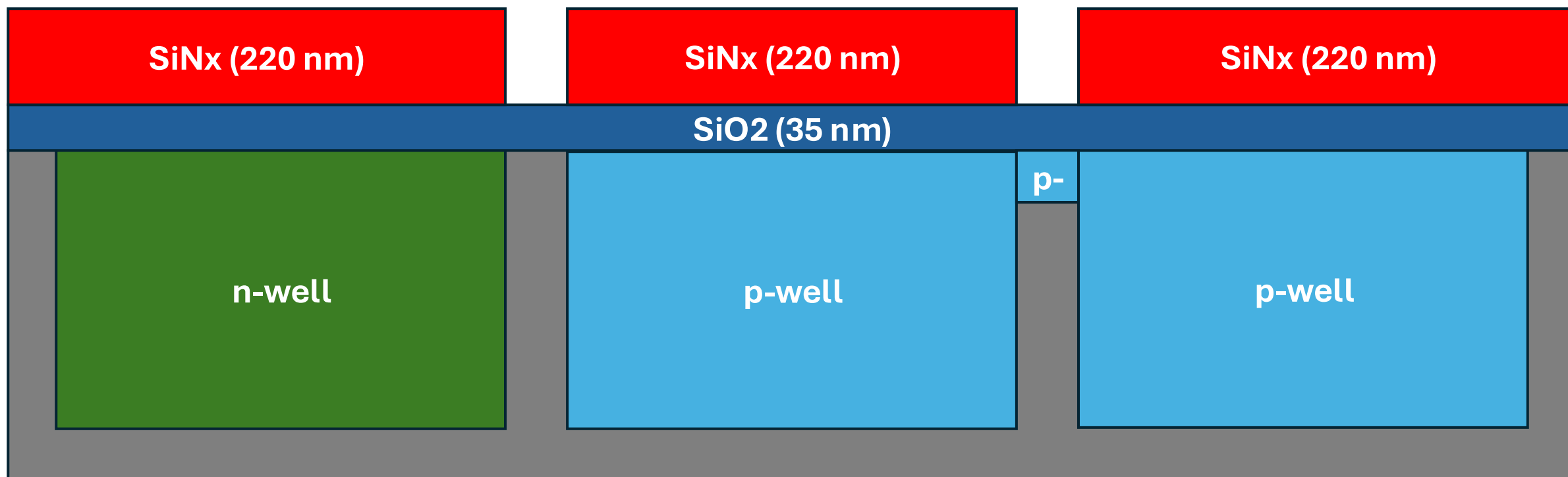


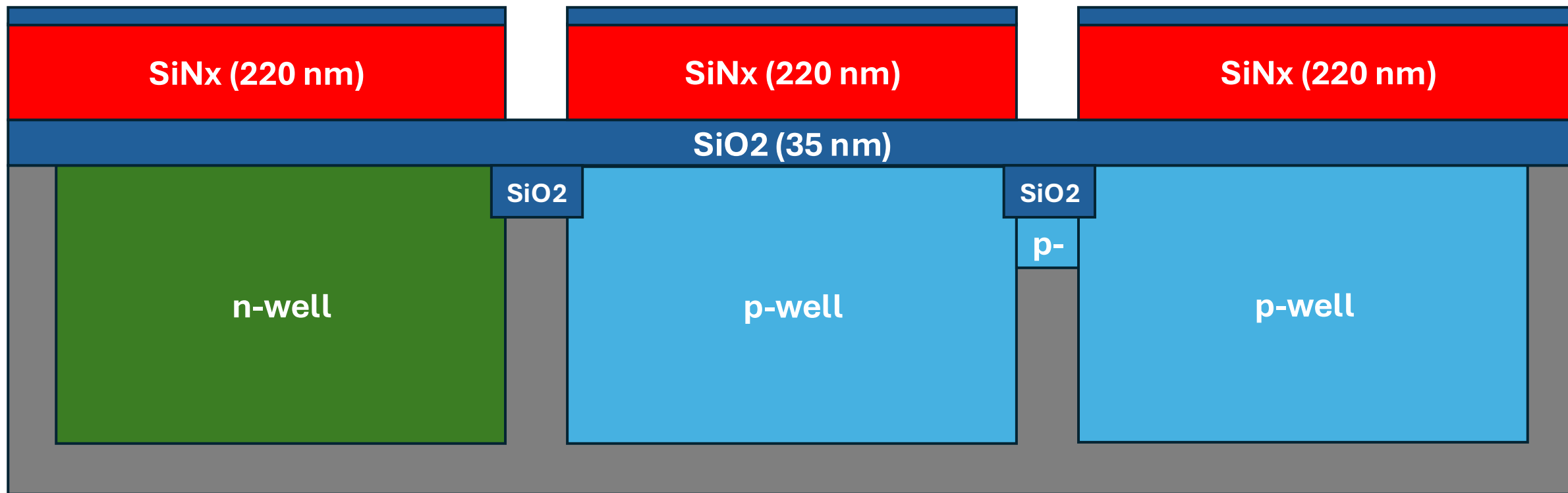


24. p-well Field Implant Litho



25. p-well Field Implant

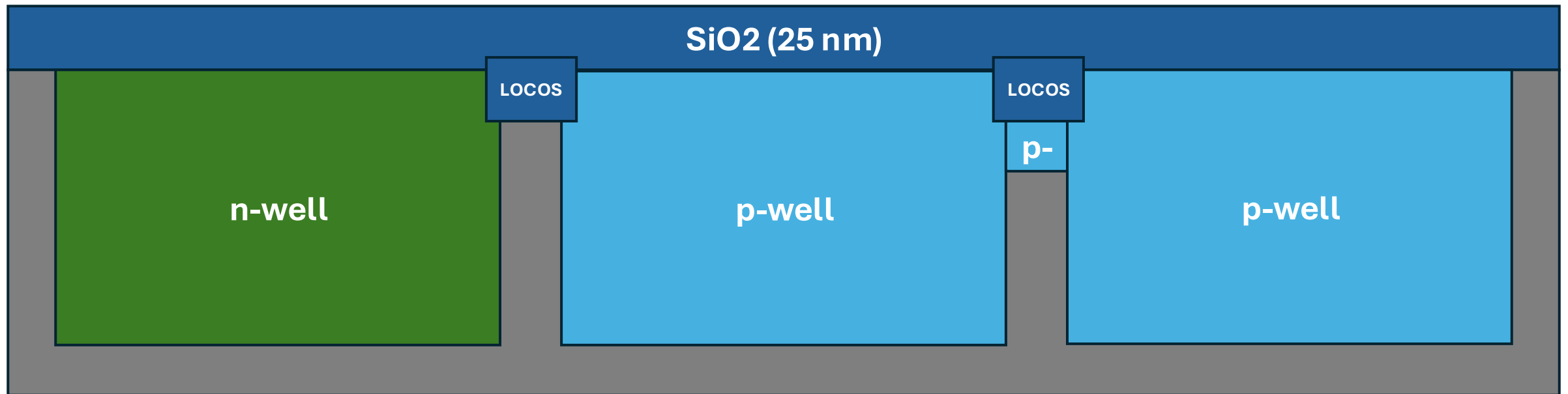




27. LOCOS (550 nm)



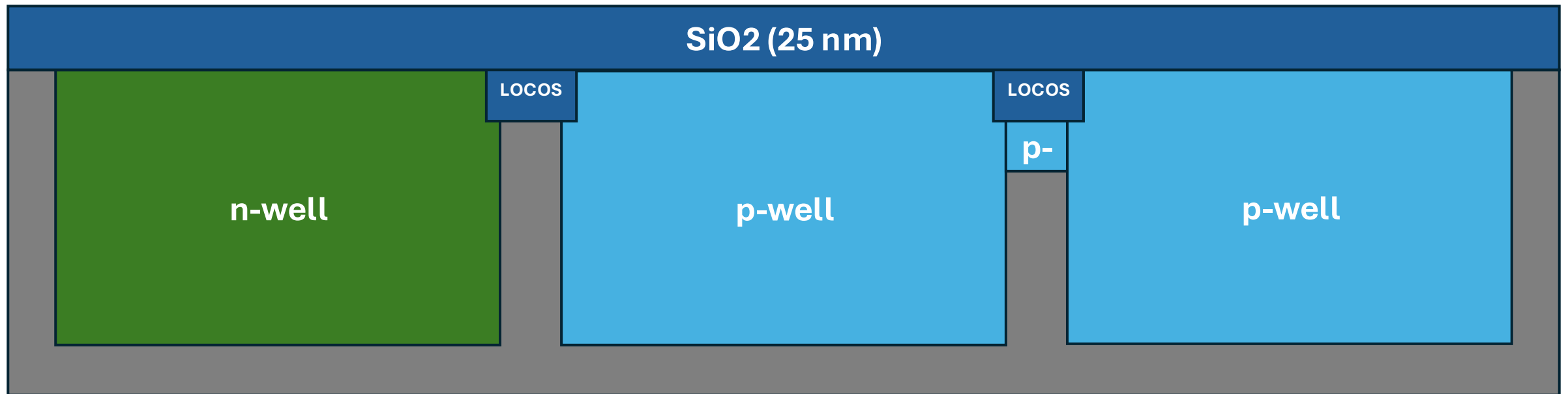
28. Thin Oxide + Nitride + Pad Oxide Removal



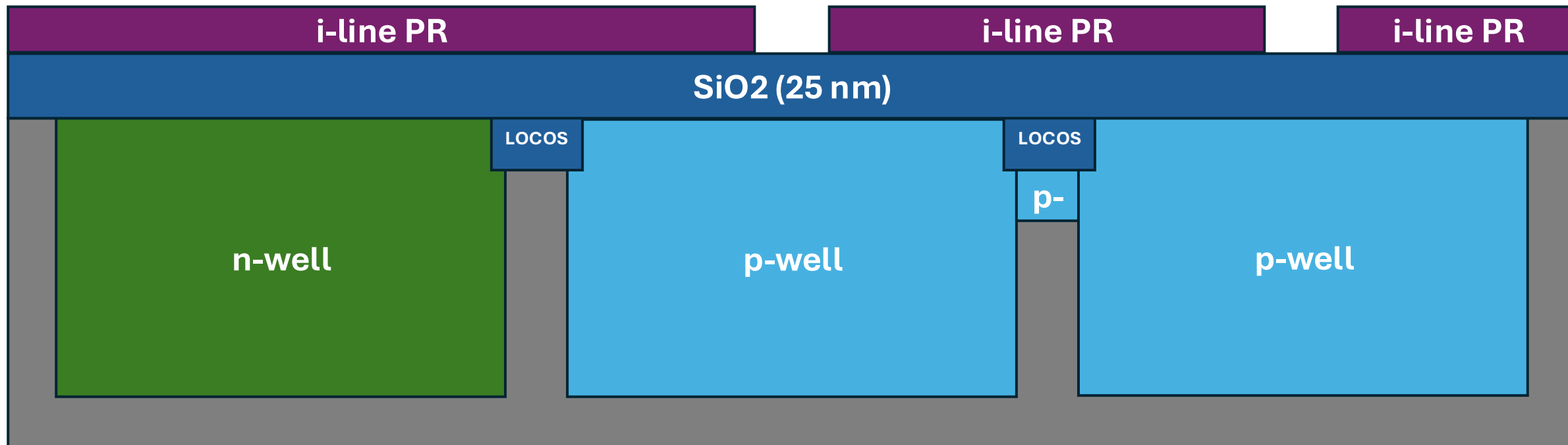
29. Sacrificial Oxidation



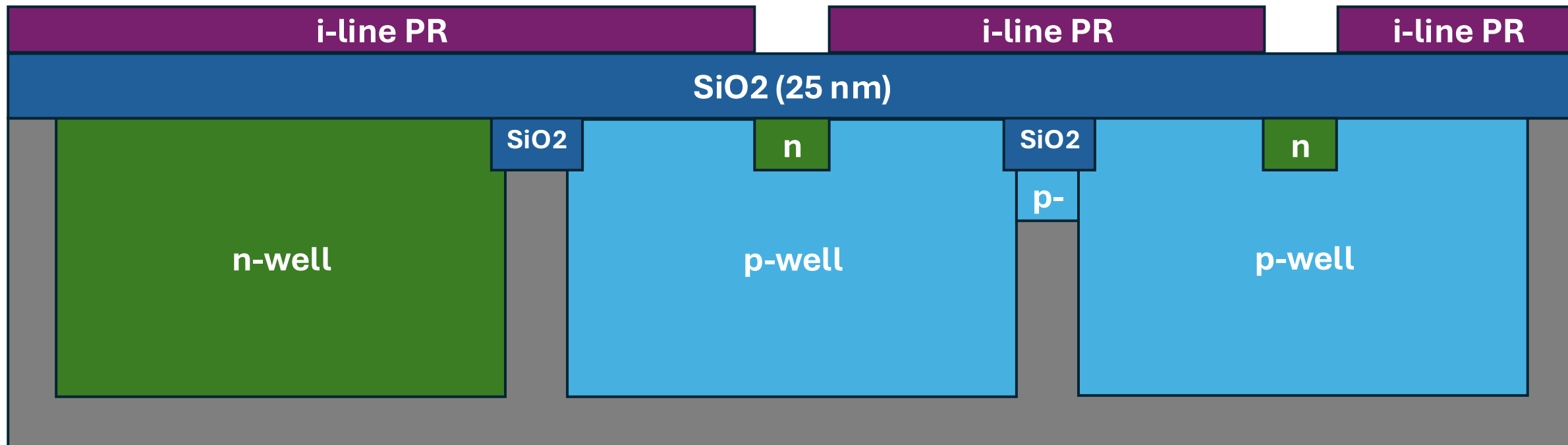
30. Sacrificial Oxide Removal



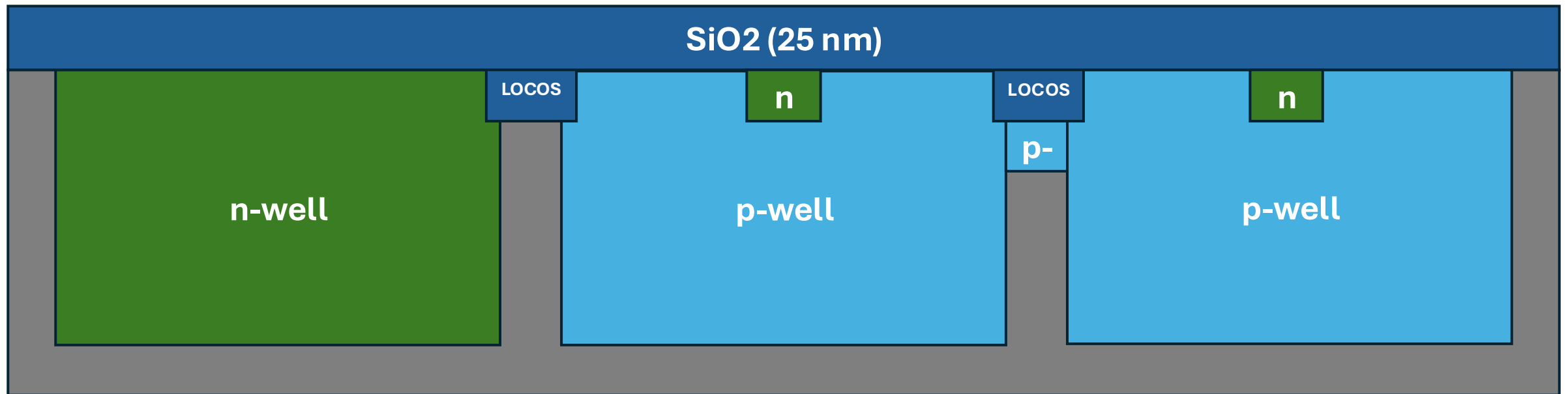
31. Screen Oxidation

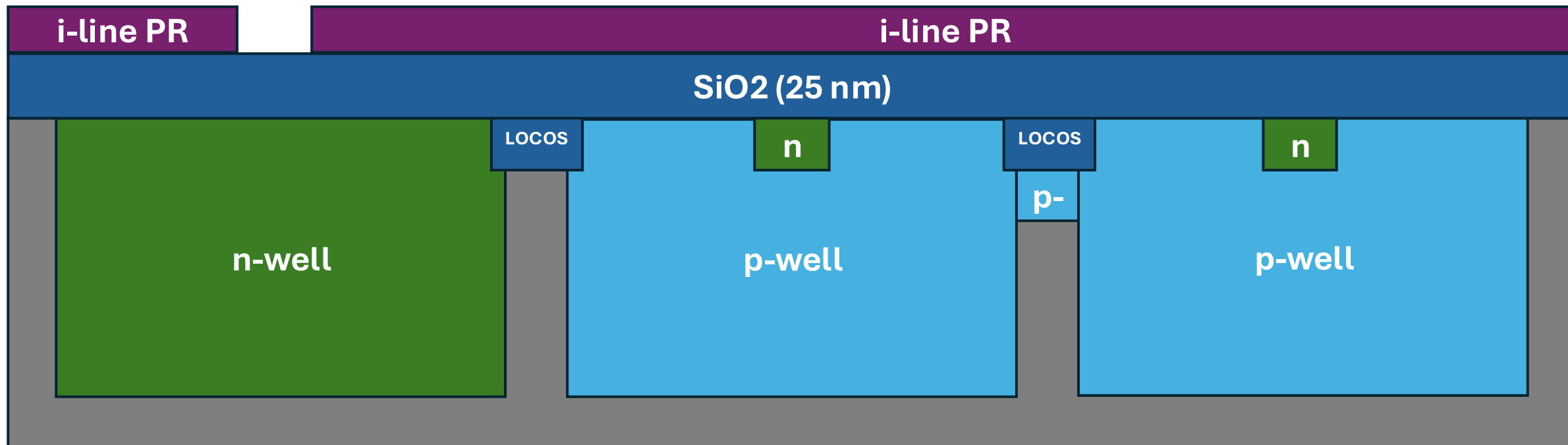


32. NMOS Vt Implant Litho

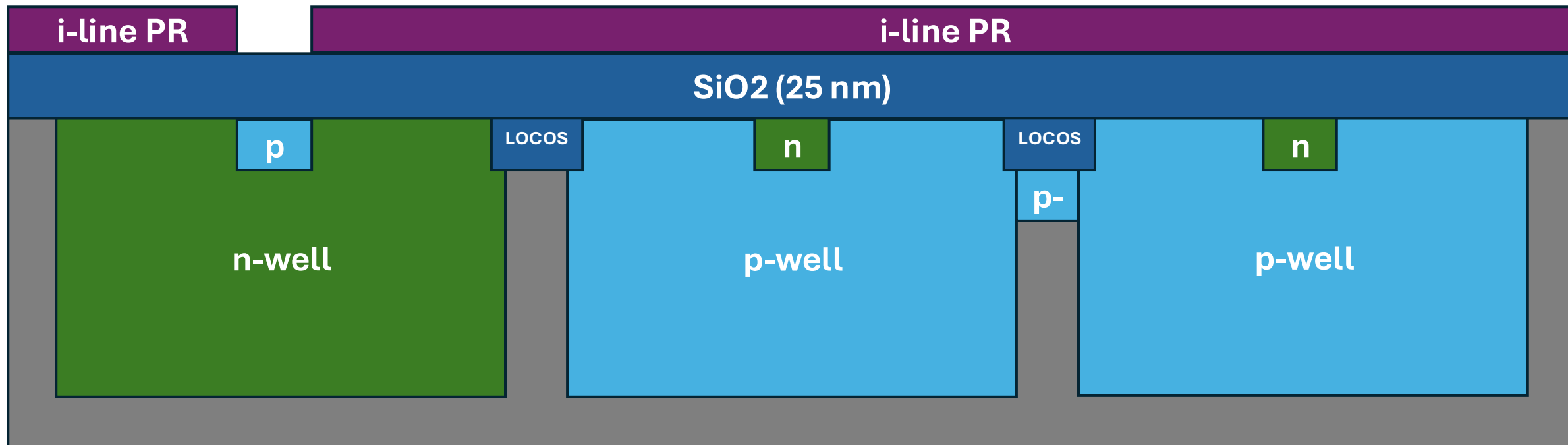


33. NMOS V_t Implant

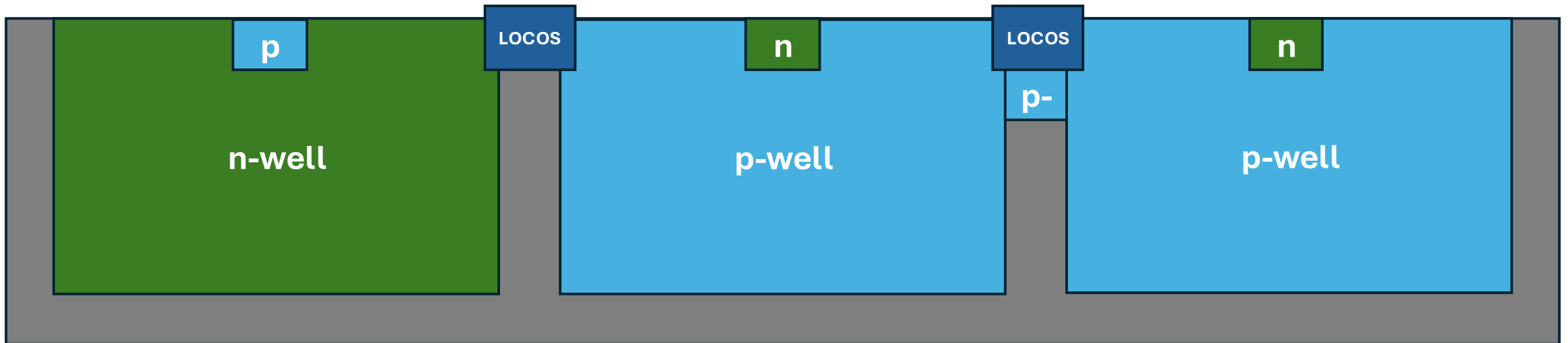




35. PMOS Vt Implant Litho



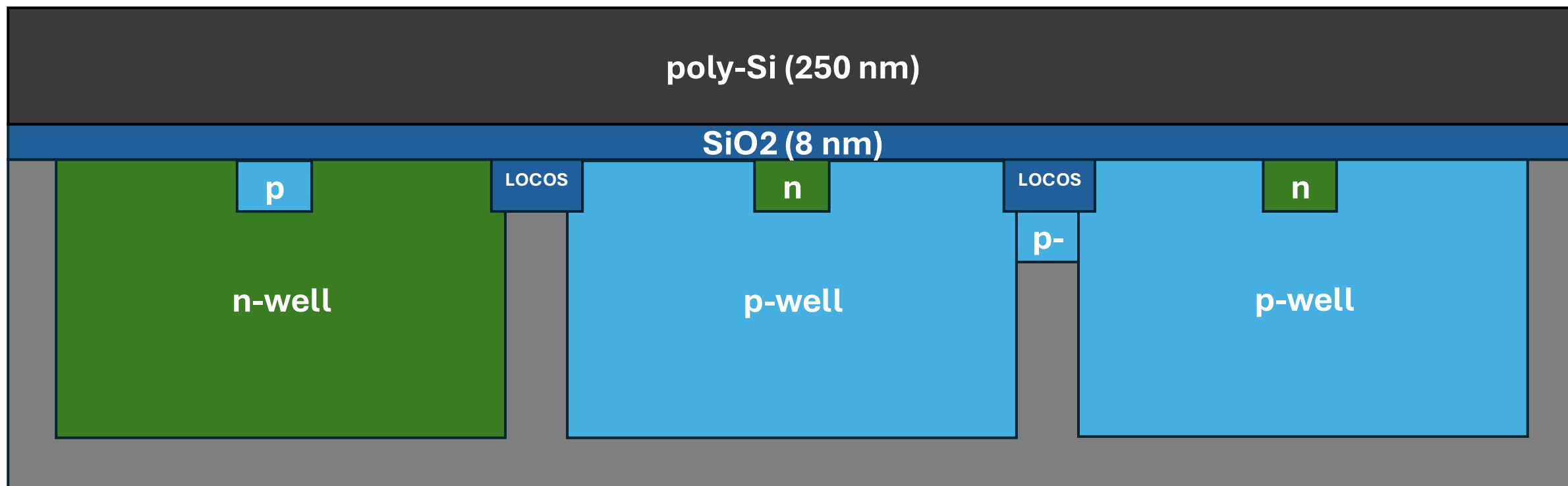
36. PMOS Vt Implant



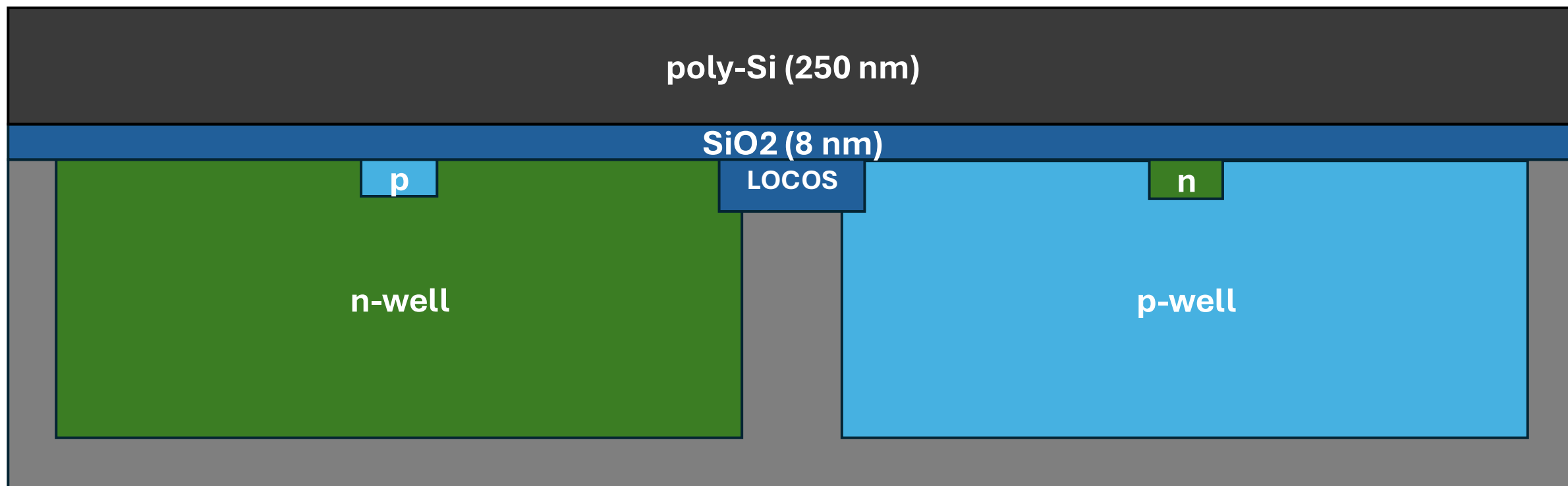
37. PR Strip + Oxide Removal



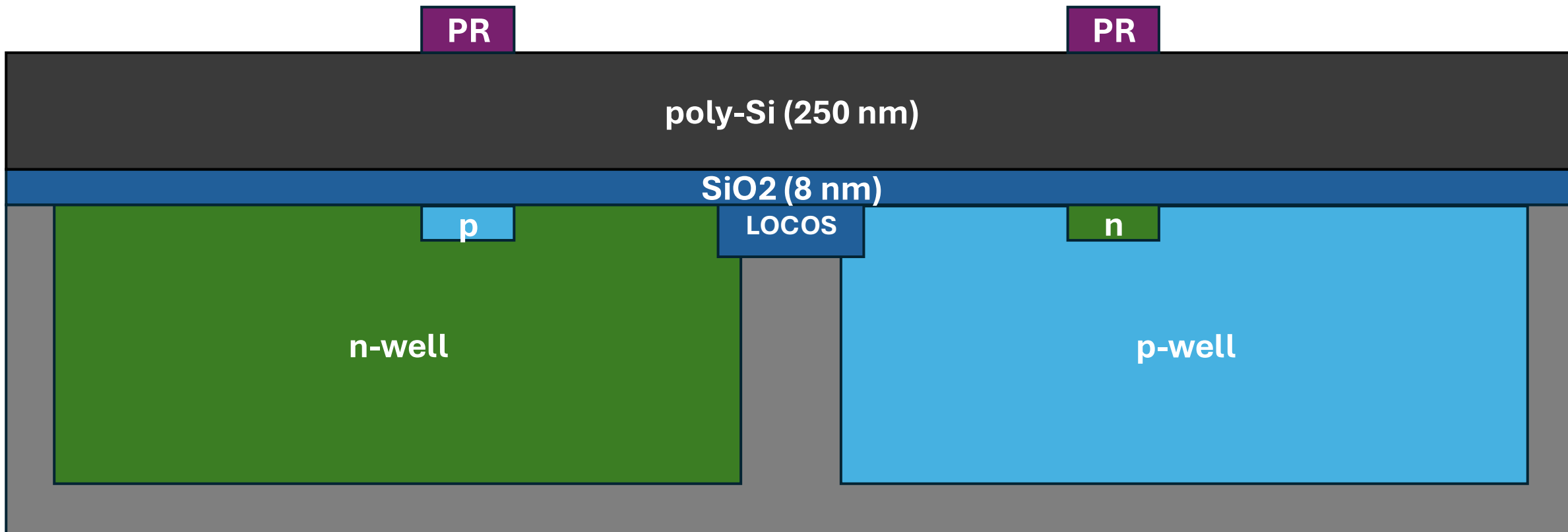
38. Gate Oxidation CVD



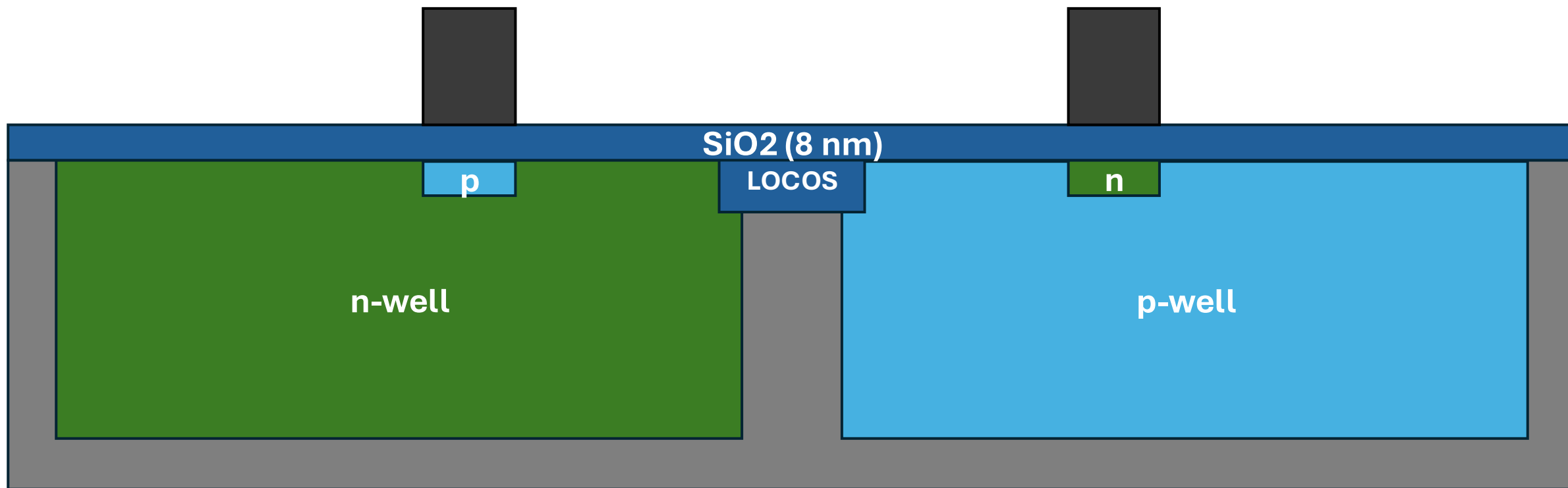
39. poly-Si CVD



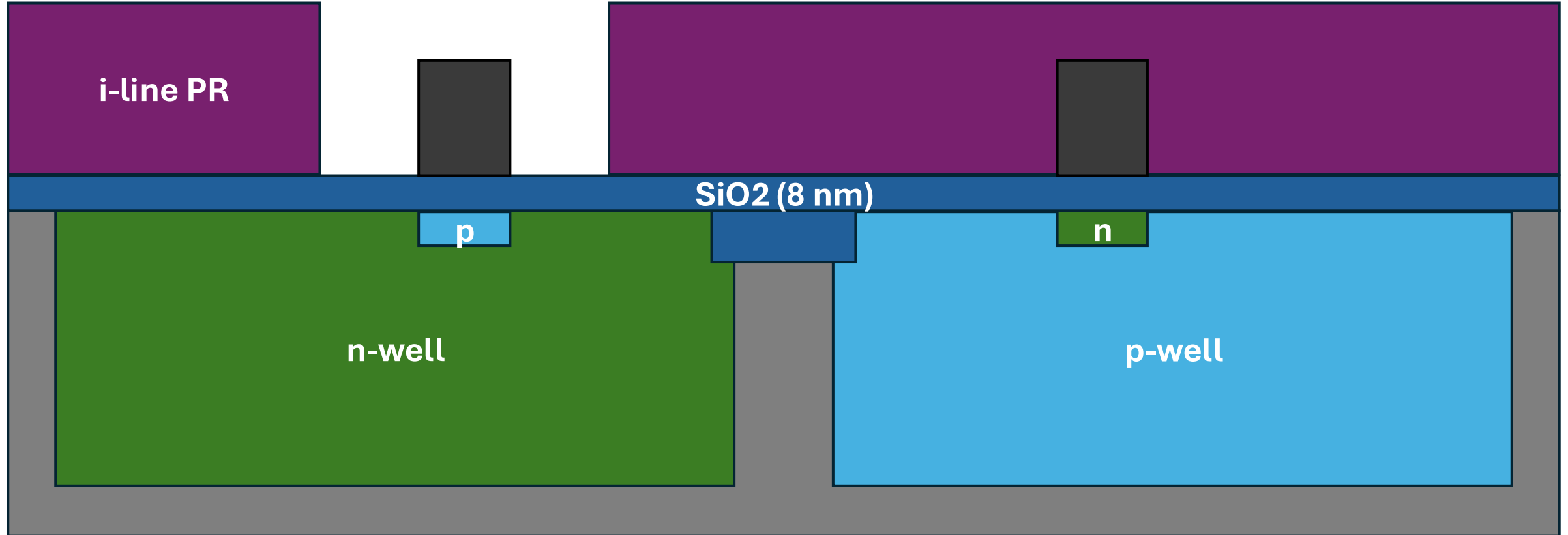
Converted to 2-wells only



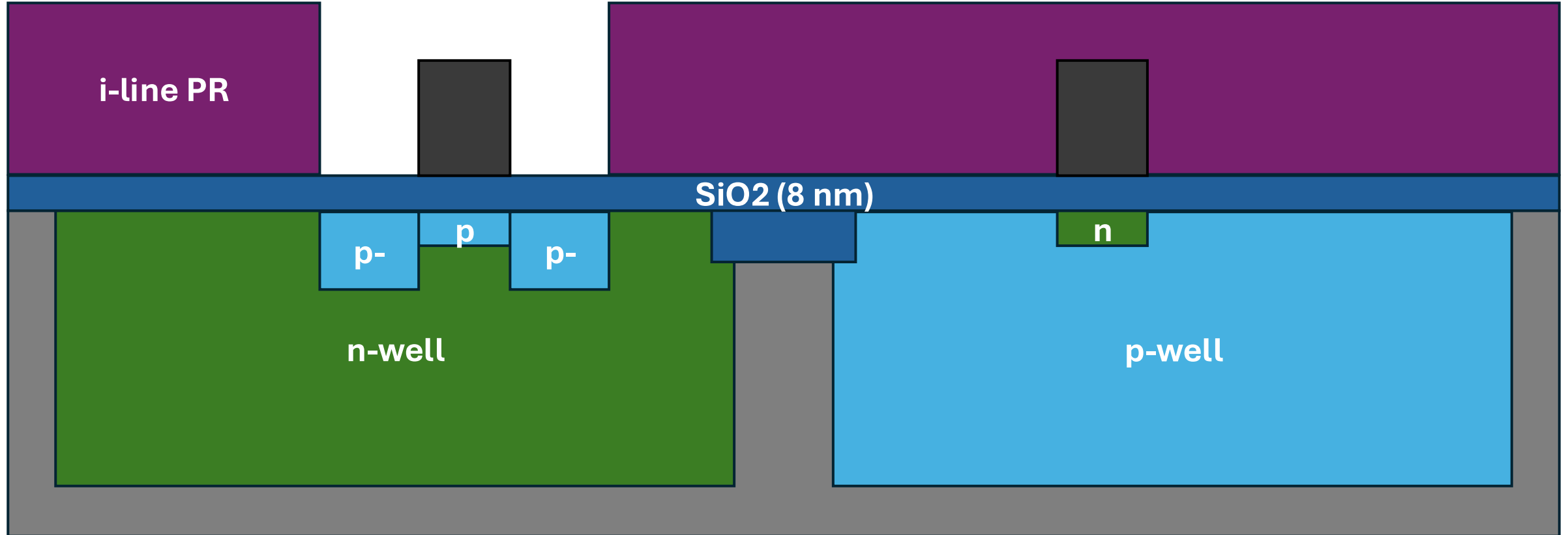
40. poly-Si Gate Litho



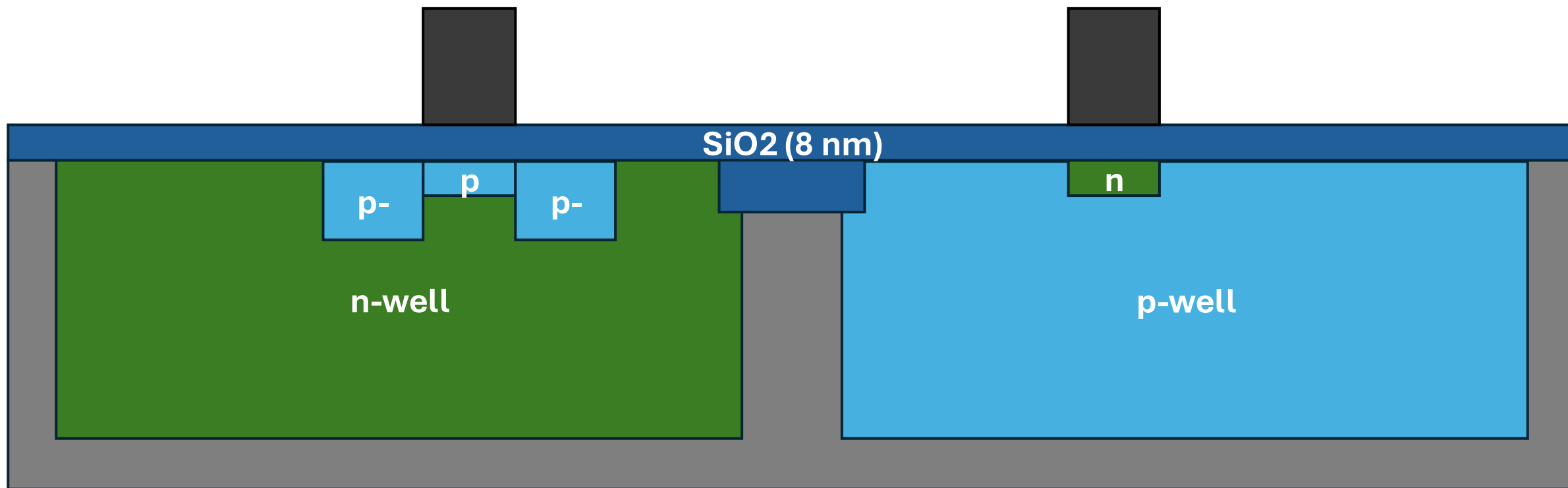
41. poly-Si Etch + PR Strip

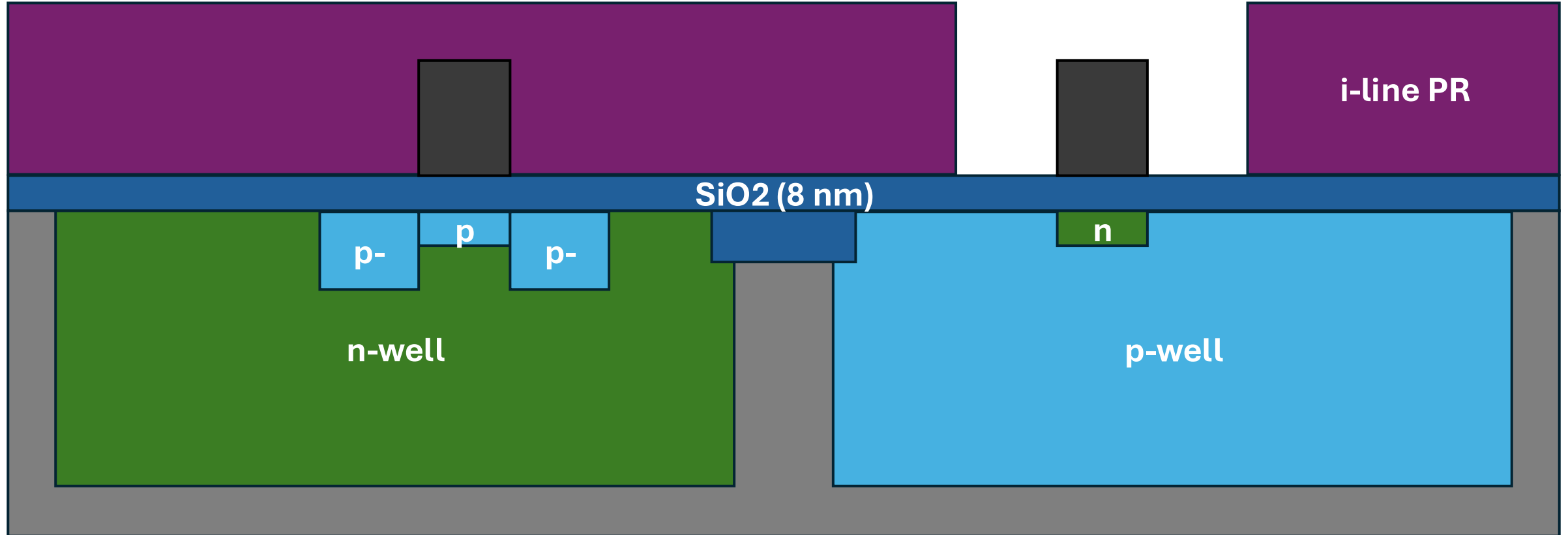


42. PMOS LDD Implant Litho

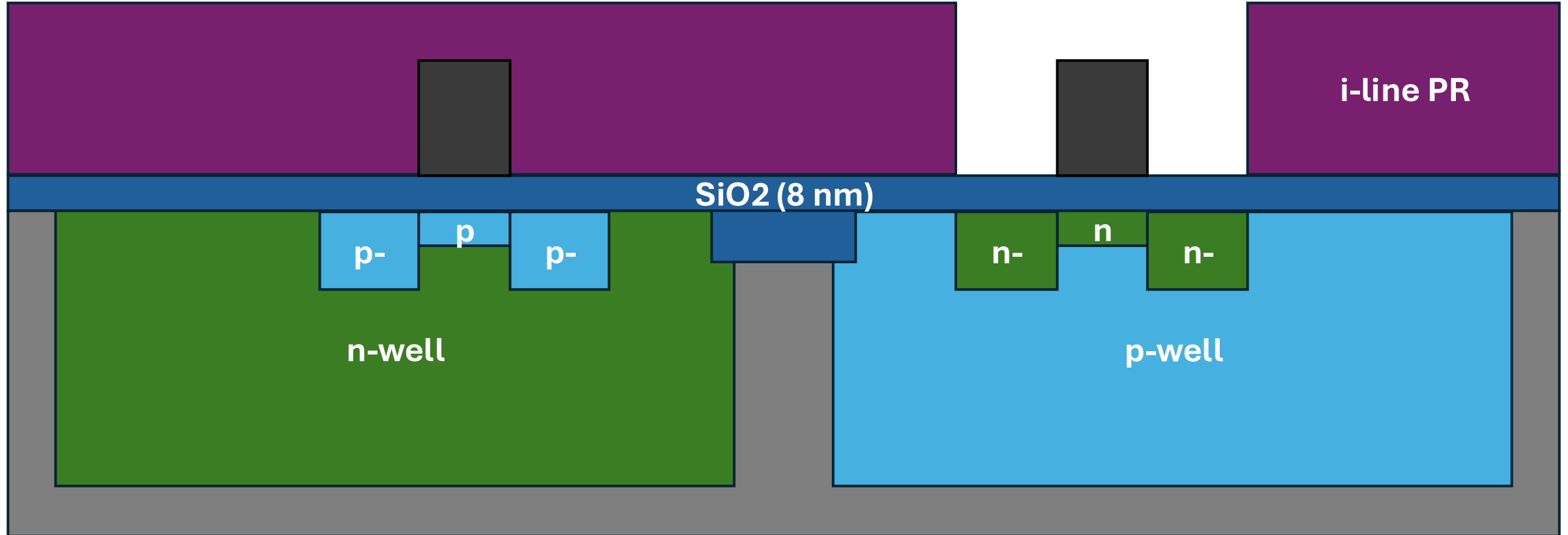


43. PMOS LDD Implant

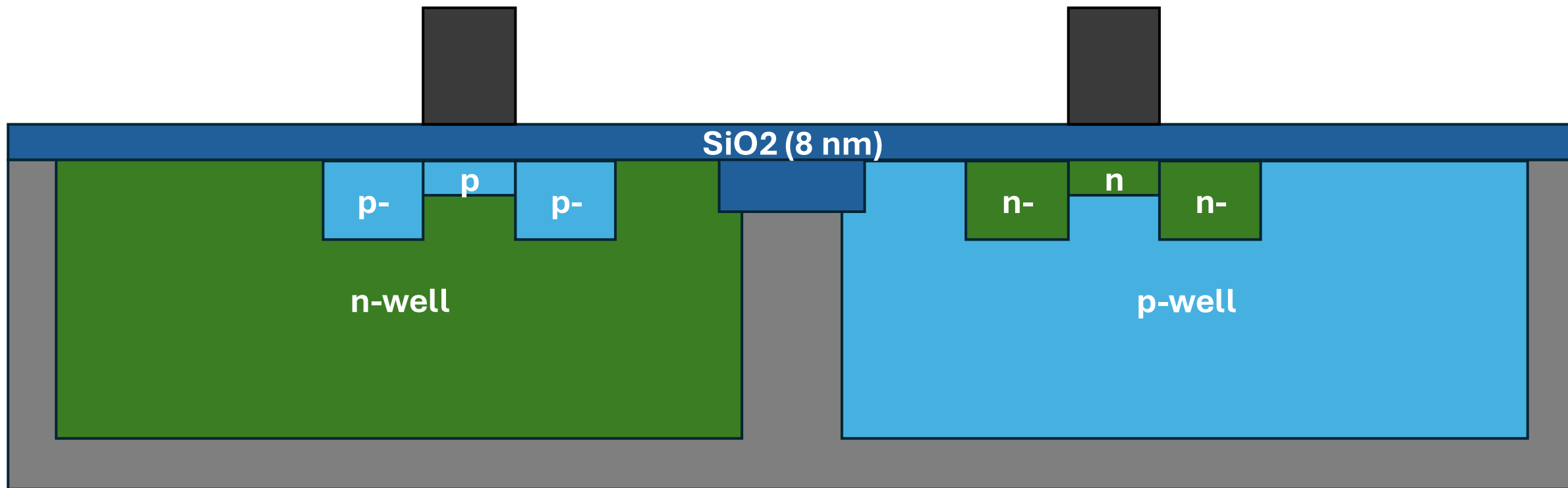


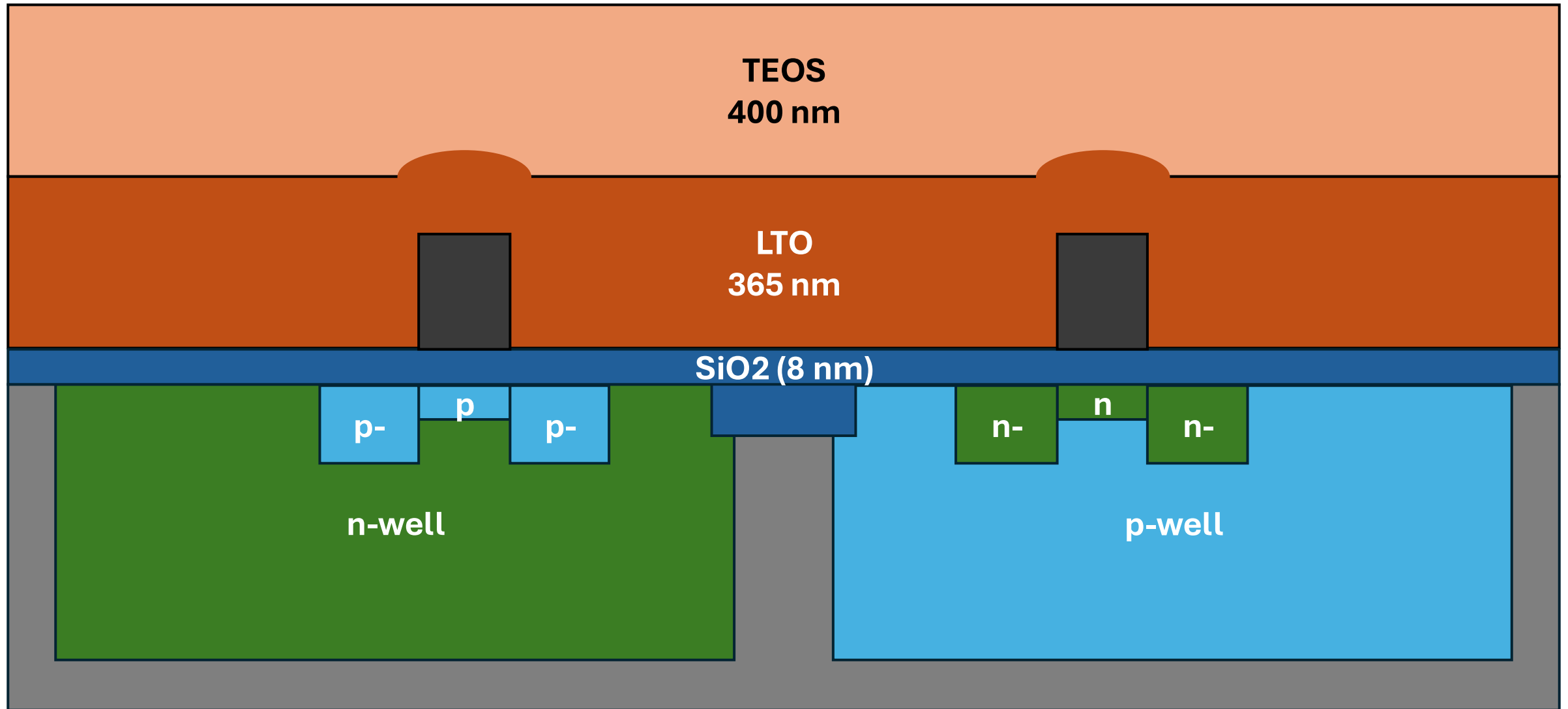


45. NMOS LDD Implant Litho

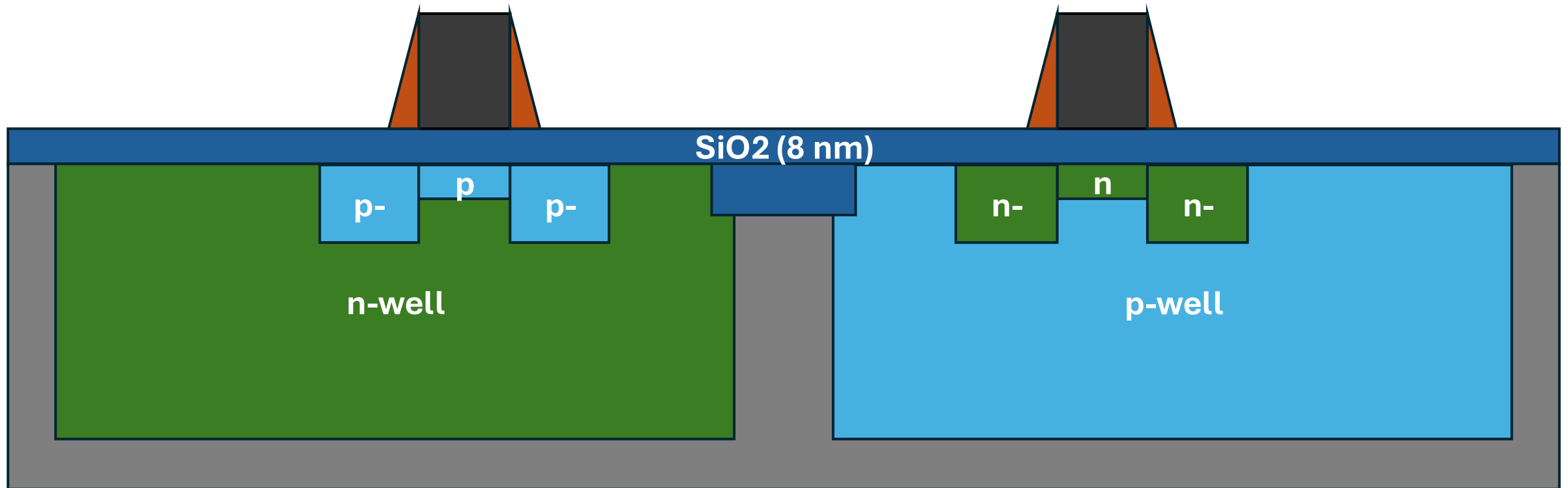


46. NMOS LDD Implant

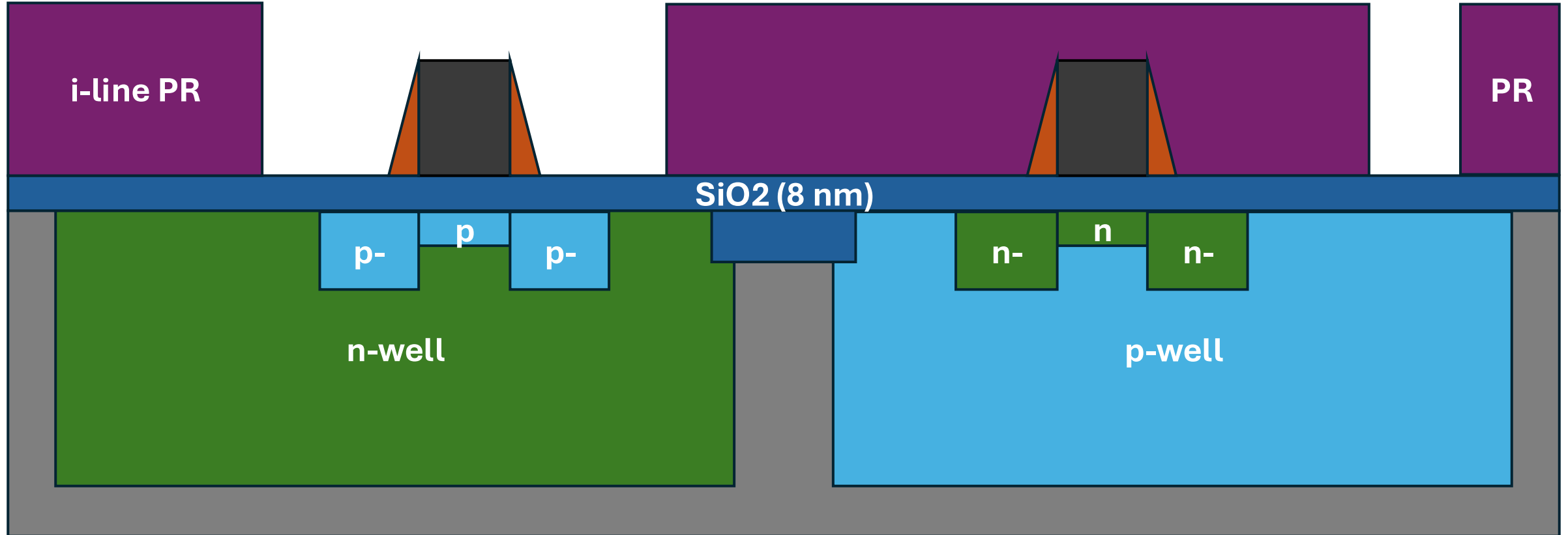




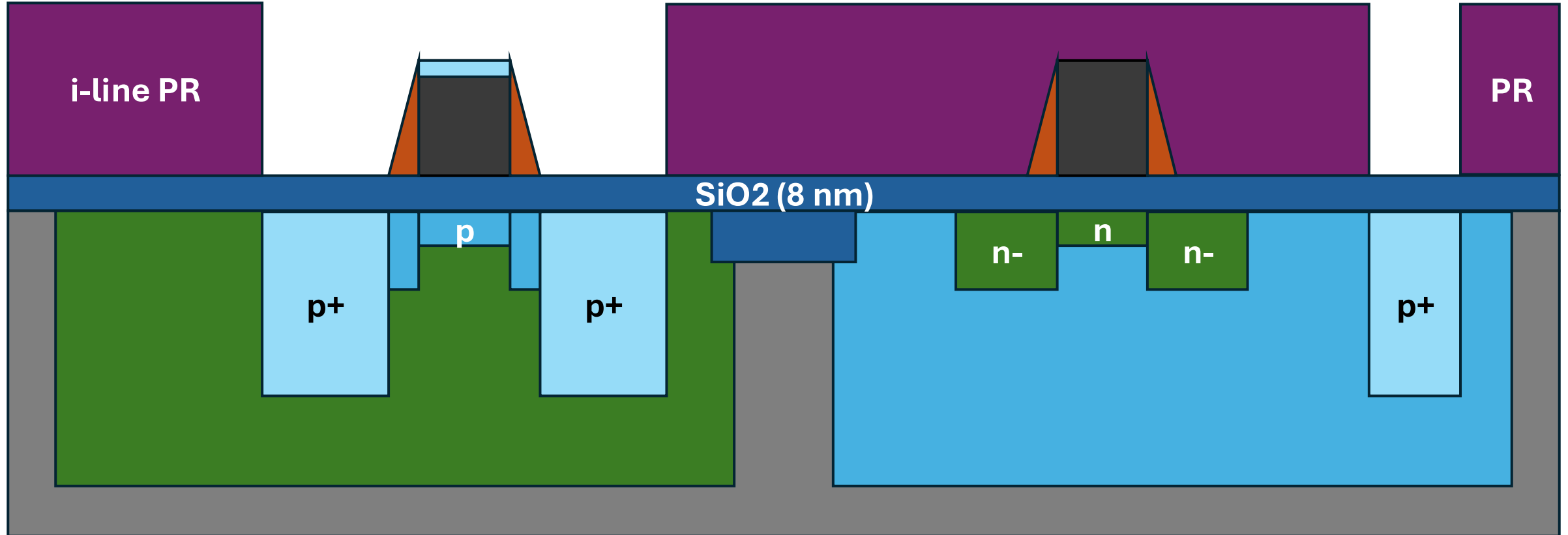
48. LTO + TEOS Deposition



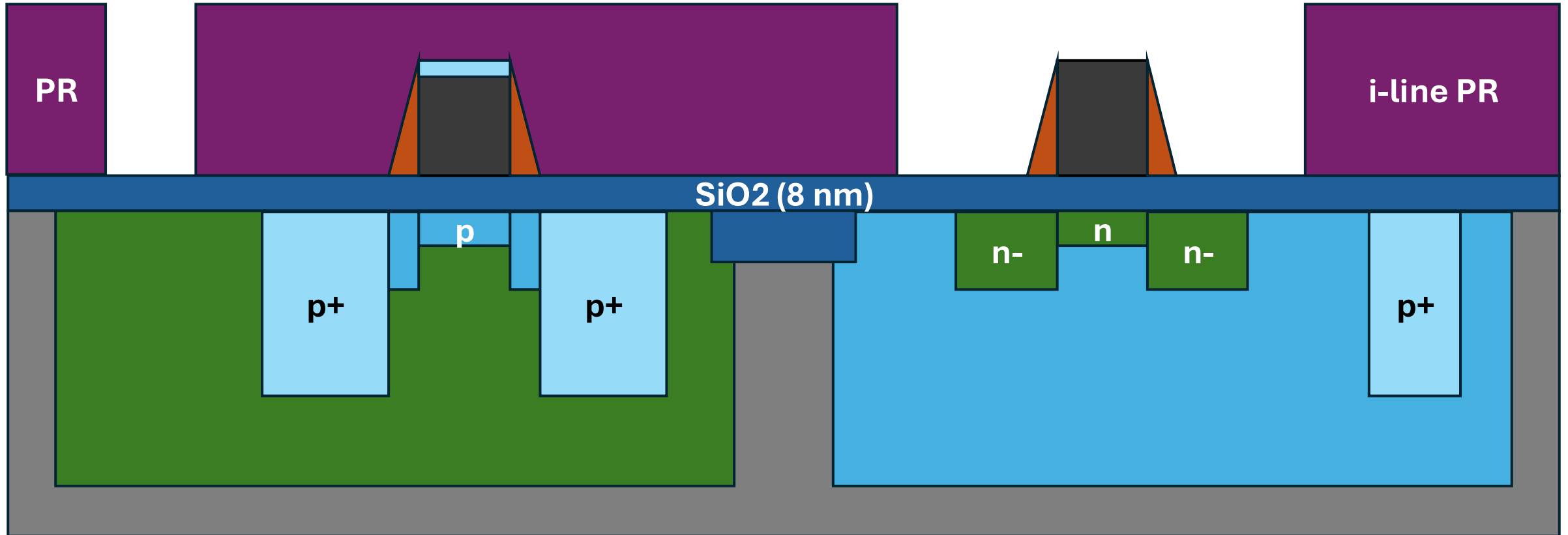
49. Spacer Formation (Blanket Etch)



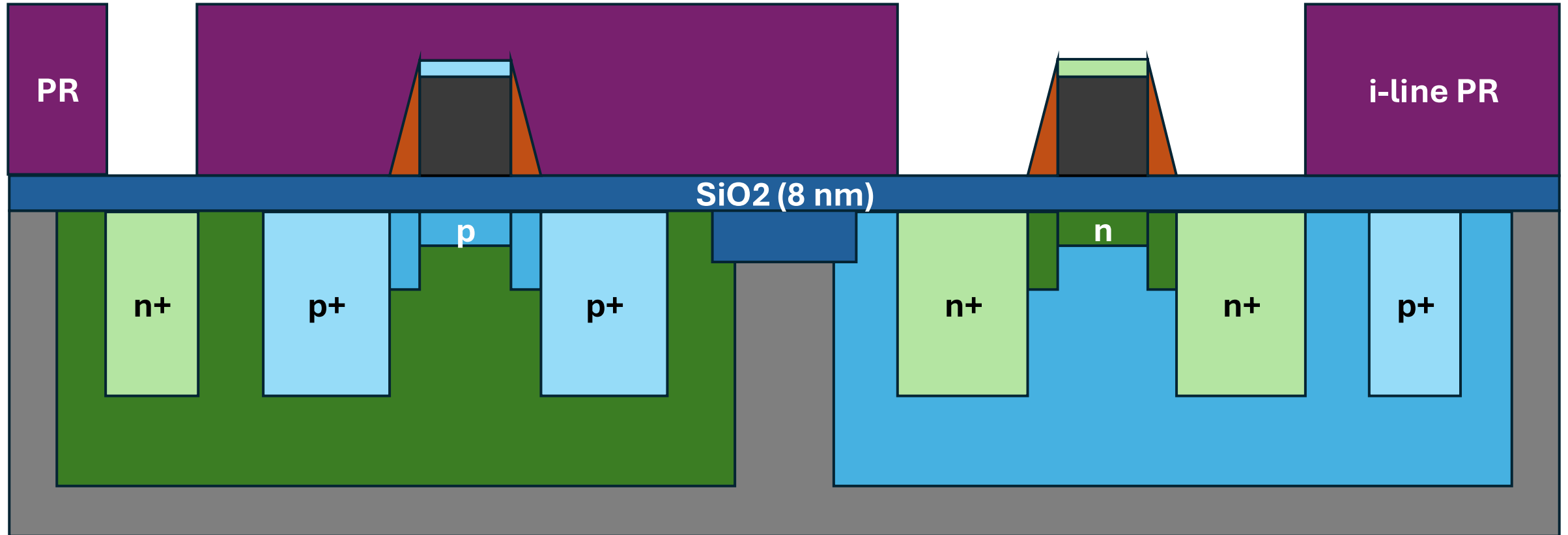
50. p+ Gate & S/D + GND Implant Litho



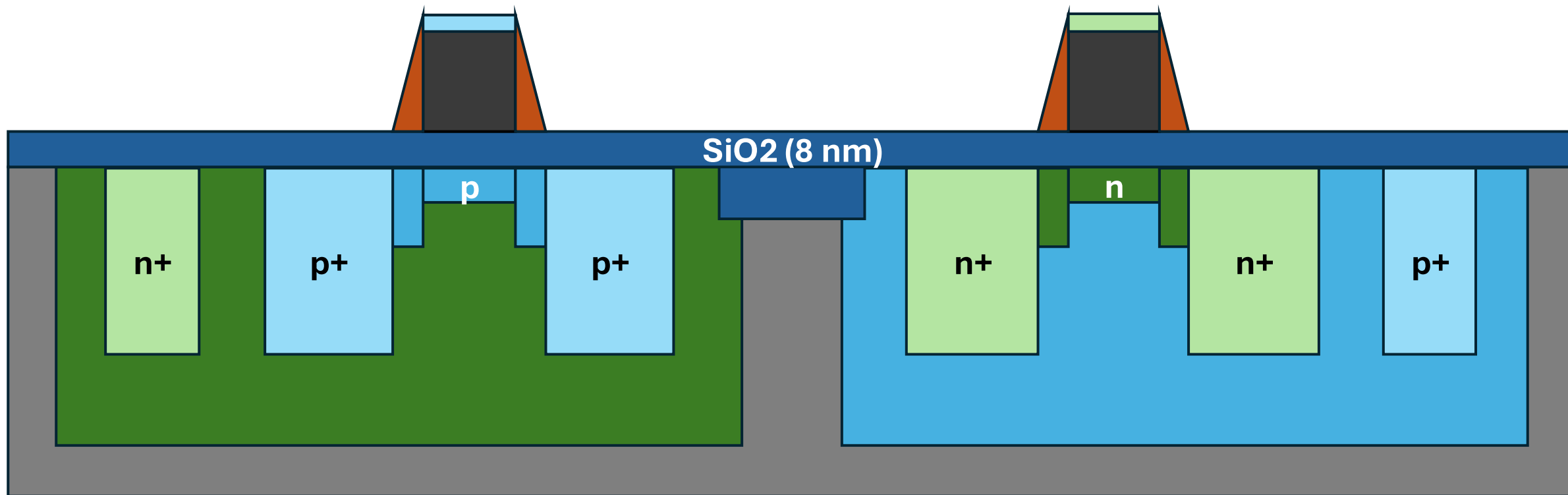
51. p+ Gate & S/D + GND Implant



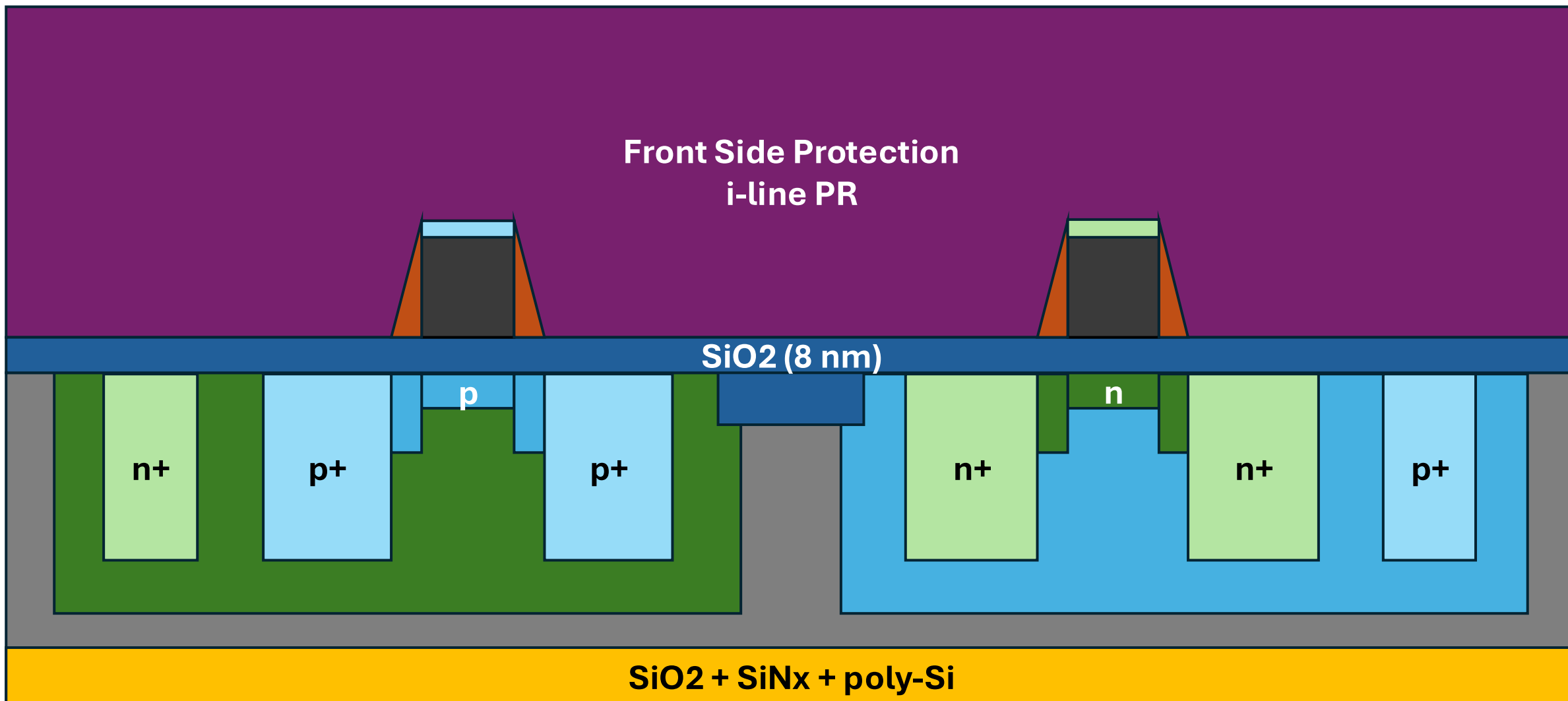
52. n+ Gate & S/D + GND Implant Litho



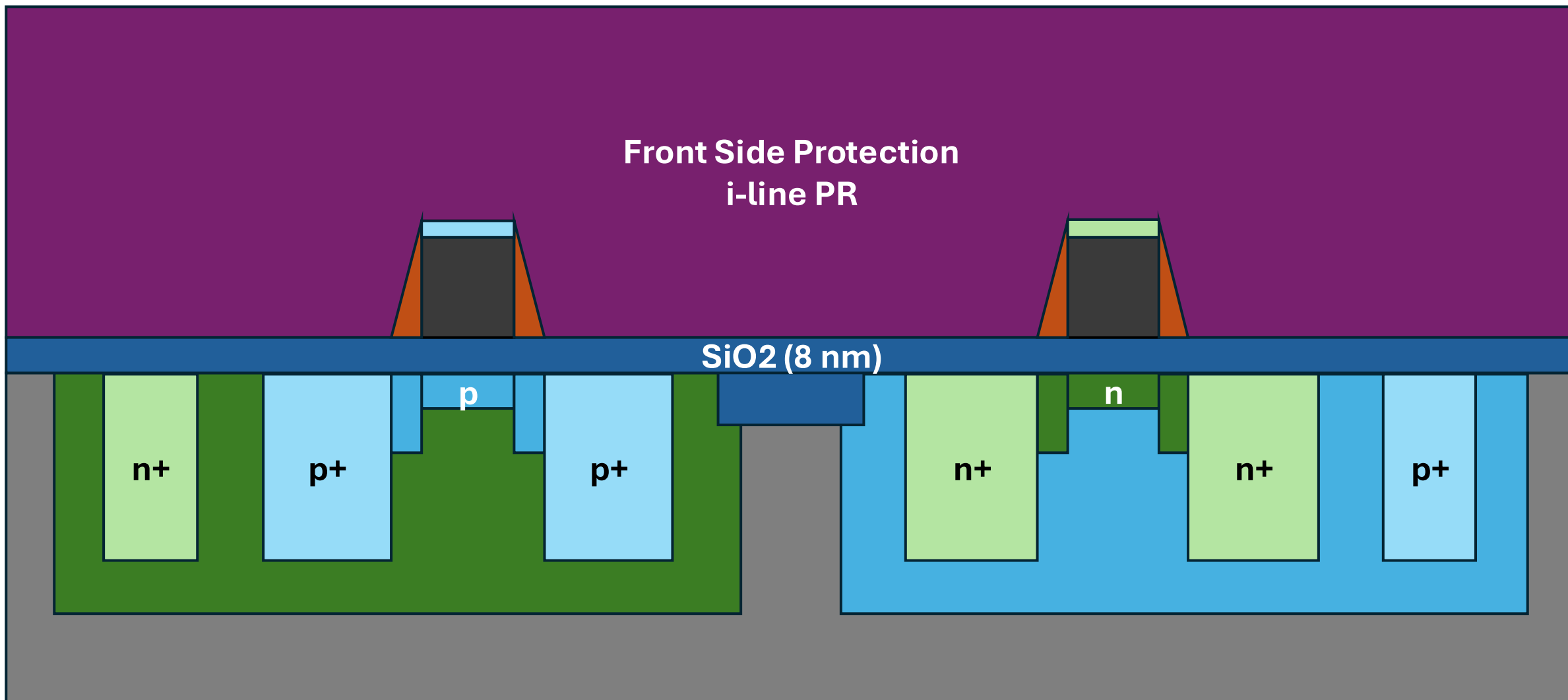
53. n+ Gate & S/D + GND Implant



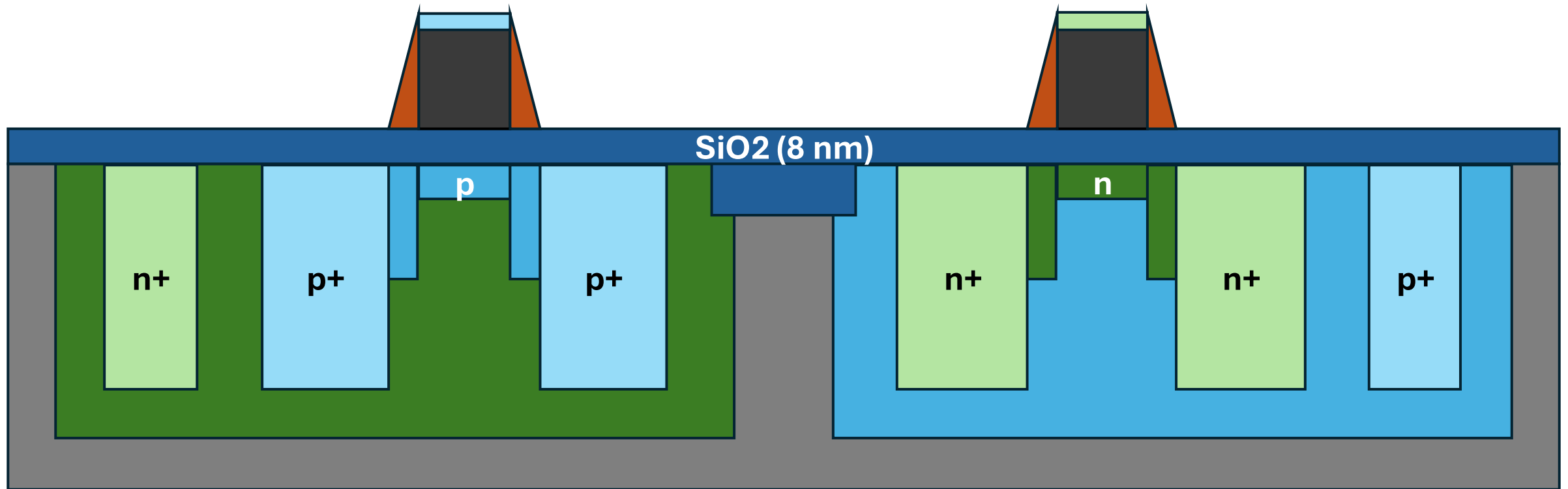
54. PR Strip



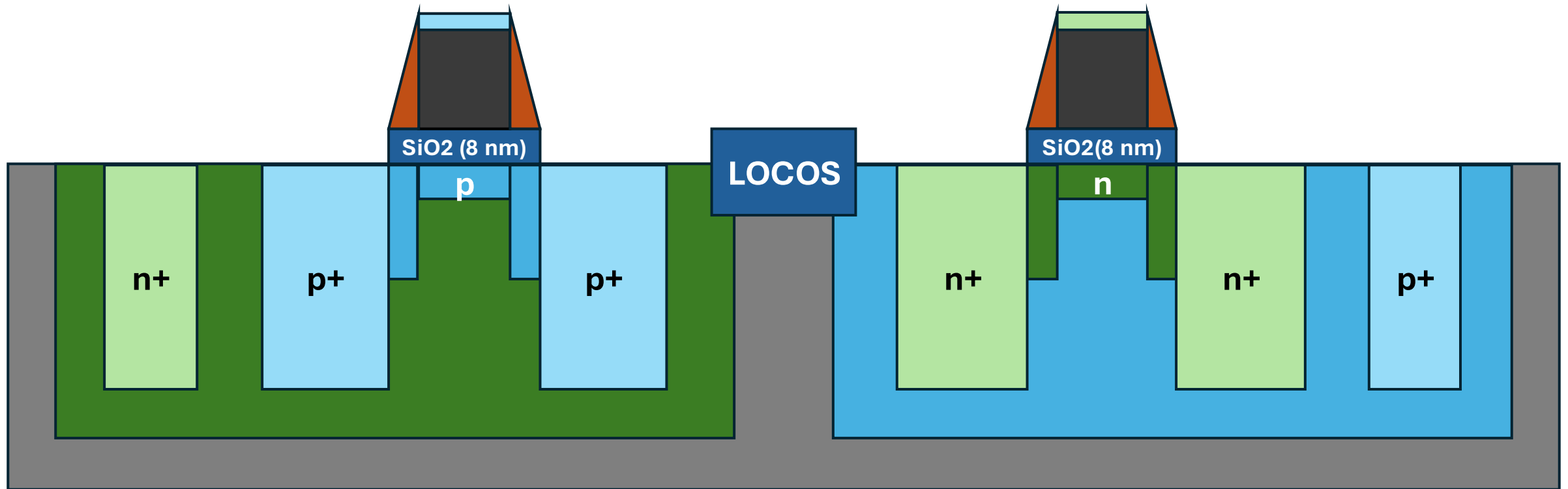
55. PR Coating for Front Side Protection



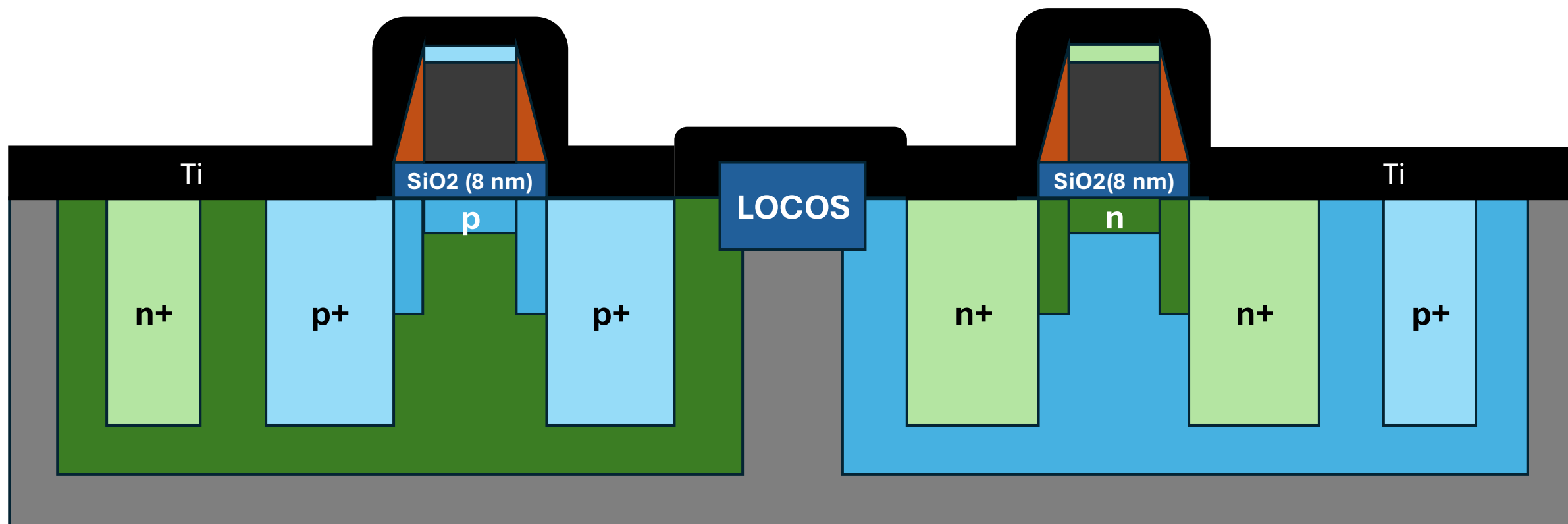
56. Oxide + Nitride + poly-Si Etch



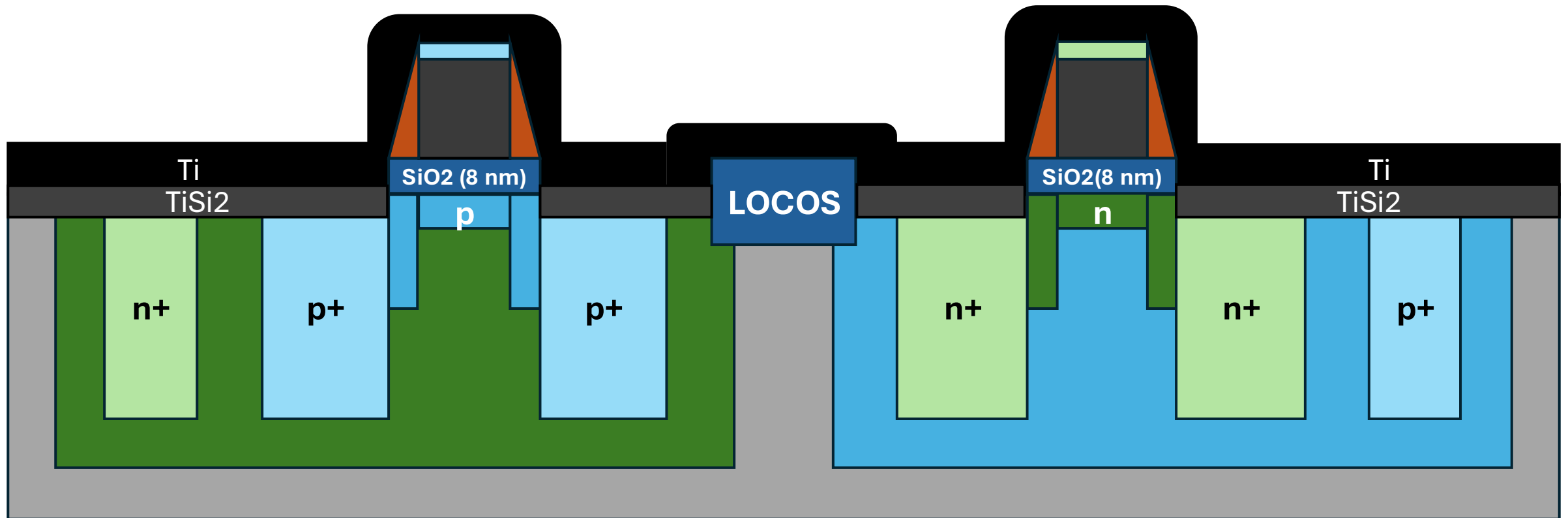
57. PR Strip + S/D Anneal



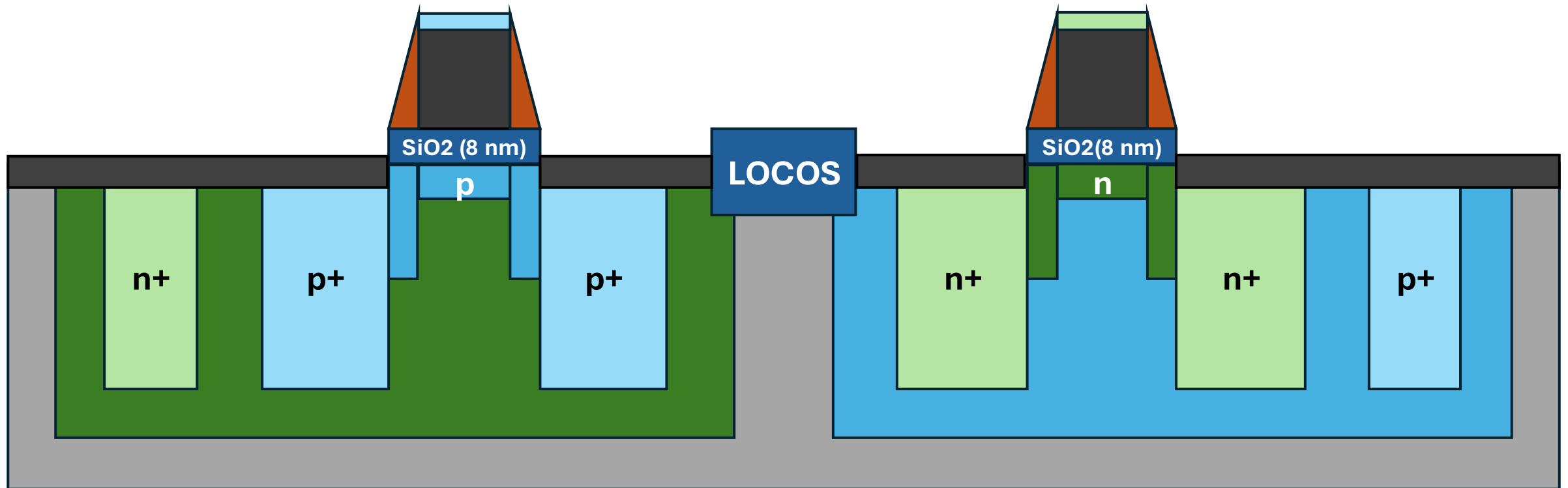
58. Sputter Etch



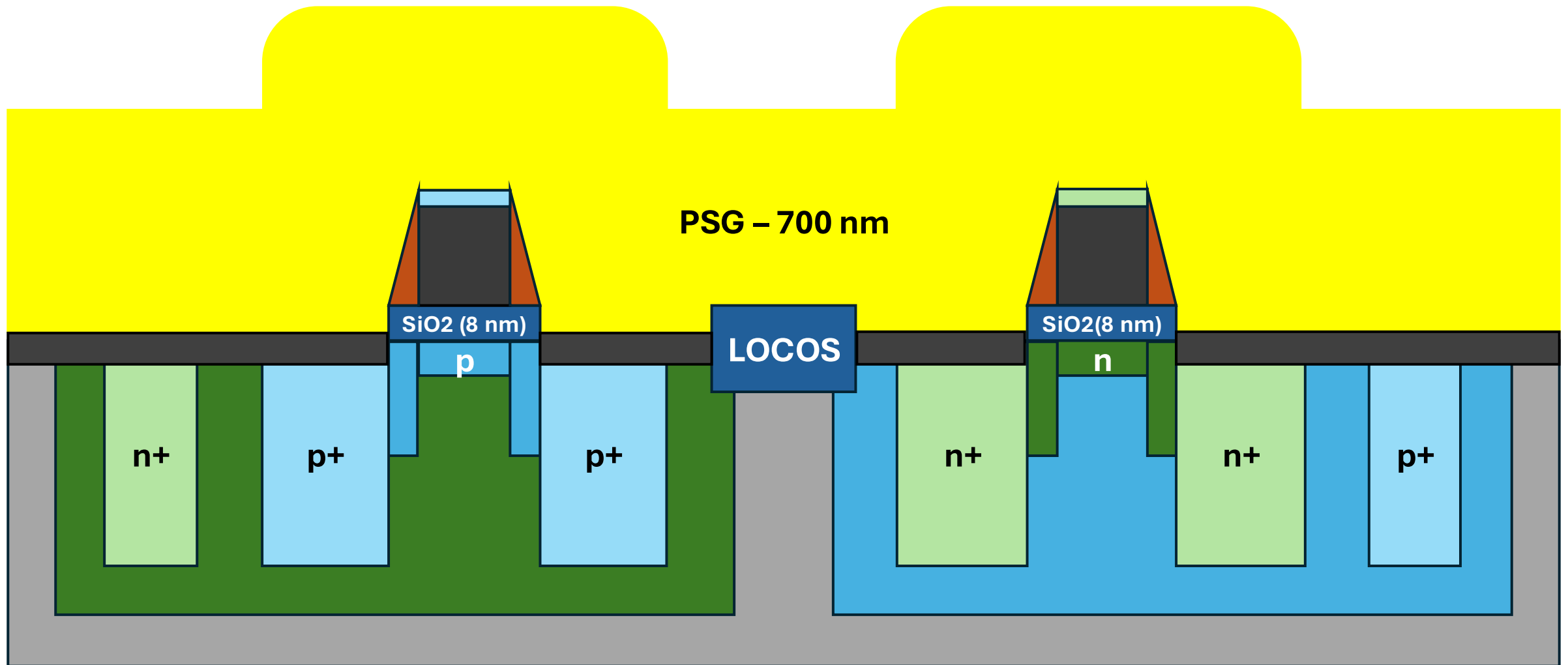
59. Ti Deposition



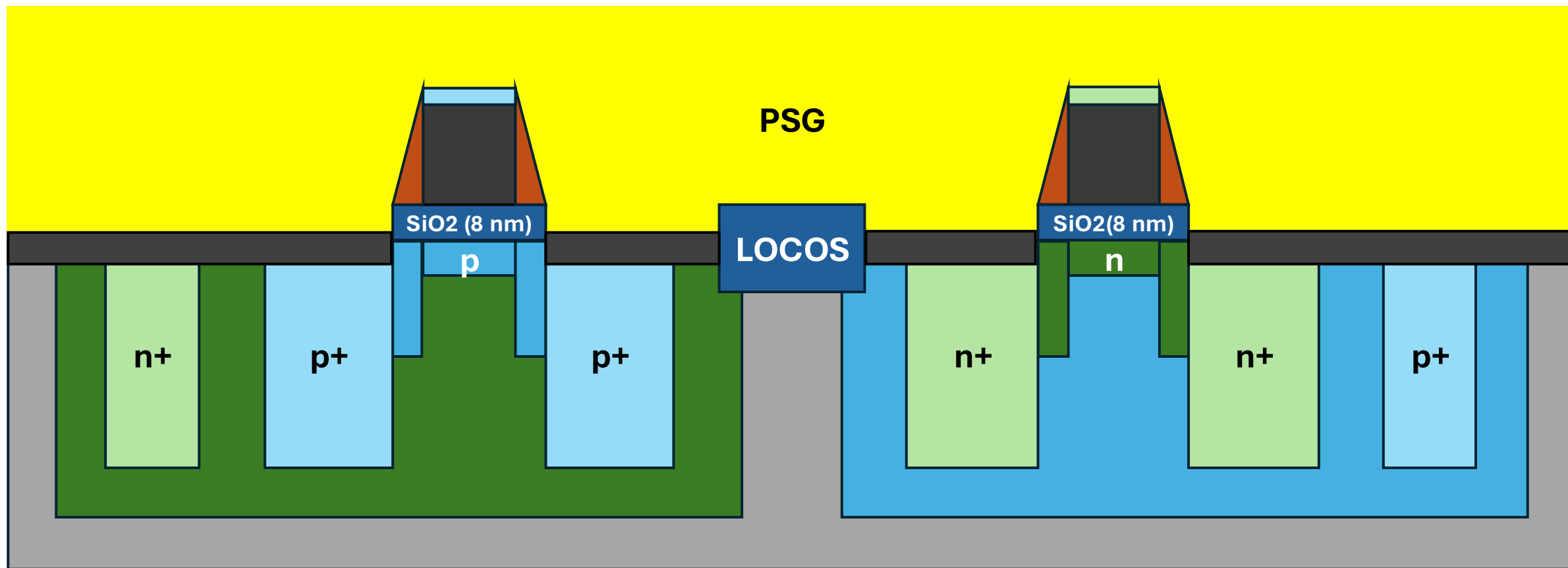
60. RTA – Silicidation (TiSi2) [N2 Chamber]

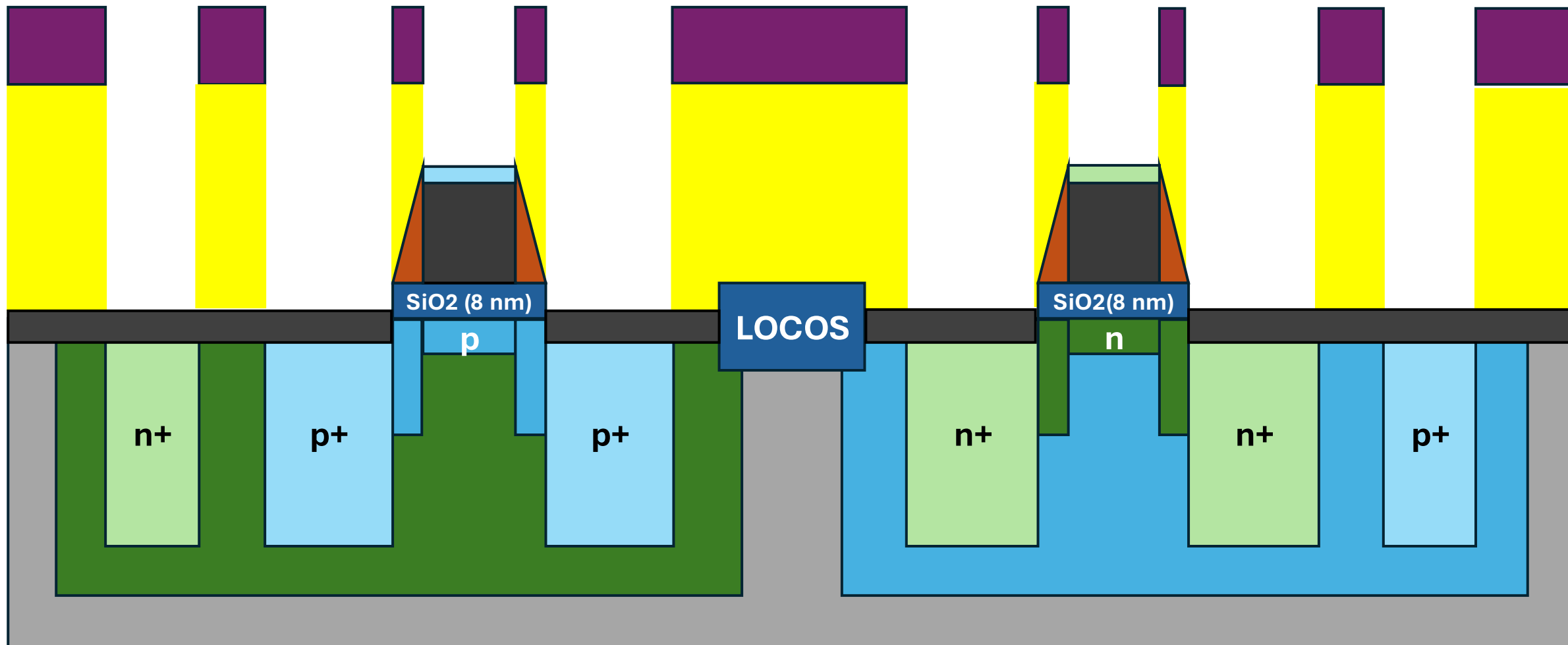


61. Wet Etch – Ti & TiN

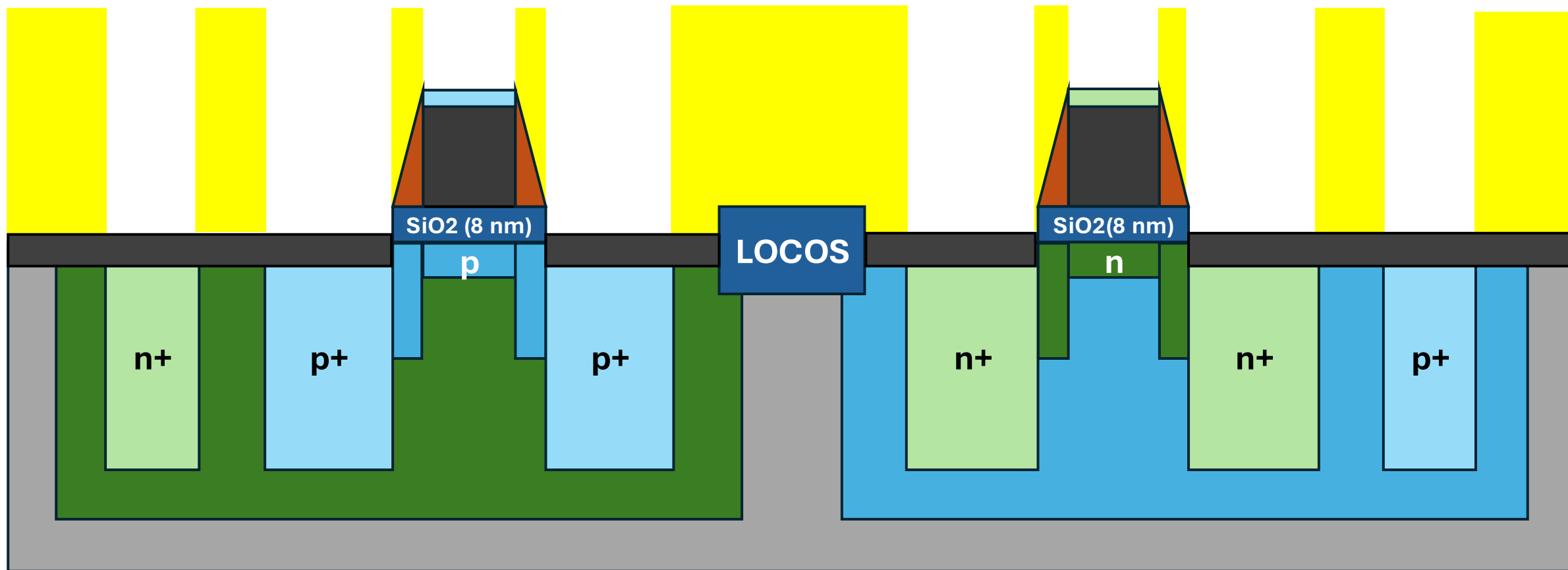


62. PSG Deposition & Densification

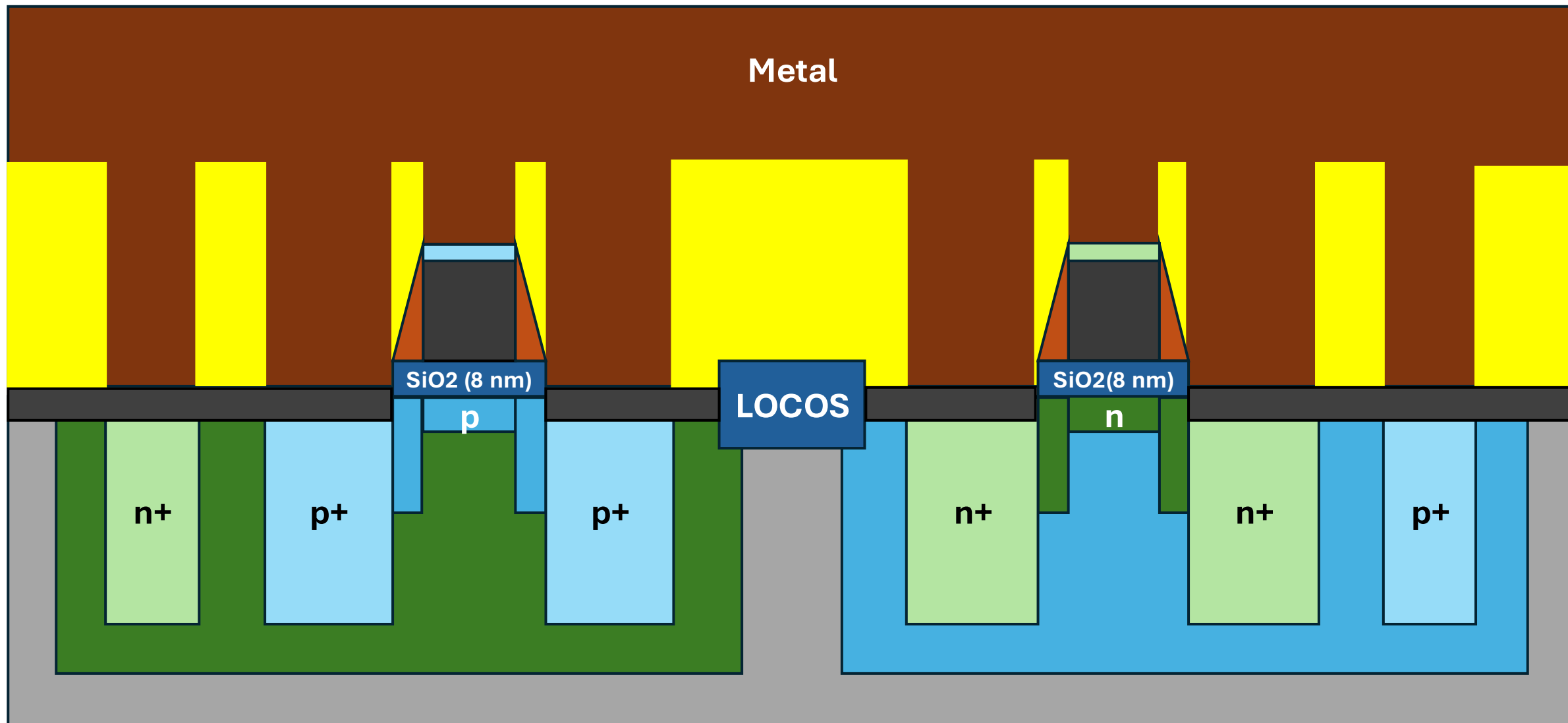




65. Contact Etch



66. PR Strip



67. Metal Deposition